Operating System (232)

35.0.1 UGC NET CSE | December 2015 | Part 3 | Question: 42

In an operating system, indivisibility of operation means:

A. Operation is interruptable
B. Race Condition may occur
C. processor can not be prempted
D. All of the above

ugcnetcse-dec2015-paper3 operating-system

Answer key

35.0.2 UGC NET CSE | December 2015 | Part 3 | Question: 41

Match the following in Unix file system :

	List-I		List-II
(a)	Boot block	(i)	Information about file system
(b)	Super block	(ii)	Information about file
(c)	Inode table	(iii)	Storage space
(d)	Data block	(iv)	Code for making OS ready

Codes :

A. (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)
B. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
C. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
D. (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)

operating-system ugcnetcse-dec2015-paper3

Answer key

35.0.3 UGC NET CSE | June 2014 | Part 2 | Question: 45

Which of the following is the correct value returned to the operating system upon the successful completion of a program ?

A. 0
B. 1
C. -1
D. Program do not return a value.

ugcnetcse-june2014-paper2 operating-system

Answer key

35.0.4 UGC NET CSE | June 2012 | Part 3 | Question: 56

____ is sometimes said to be object oriented, because the only way to manipulate kernel objects is by invoking methods on their handles.

A. Windows NT
B. Windows XP
C. Windows VISTA
D. Windows 95/98

ugcnetcse-june2012-paper3 operating-system

Answer key

35.0.5 UGC NET CSE | June 2014 | Part 3 | Question: 24

Match the following :

	List – I		List – II
a.	Timeout ordering protocol	i.	Wait for graph
b.	Deadlock prevention	ii.	Roll back
c.	Deadlock detection	iii.	Wait-die scheme
d.	Deadlock recovery	iv.	Thomas Write Rule

Codes :

A. a-iv; b-iii; c-i; d-ii
B. a-iii; b-ii; c-iv; d-i
C. a-ii; b-i; c-iv; d-iii
D. a-iii; b-i; c-iv; d-iii

ugcnetjune2014iii operating-system

Answer key

35.0.6 UGC NET CSE | December 2012 | Part 3 | Question: 3

Match the following :

	List-I		List-II
a.	Critical region	i.	Hoares Monitor
b.	Wait/signal	ii.	Mutual Exclusion
c.	Working set	iii.	Principal of locality
d.	Dead lock	iv.	Circular wait

Codes :

A. a-ii, b-i, c-iii, d-iv
B. a-i, b-ii, c-iv, d-iii
C. a-ii, b-iii, c-i, d-iv
D. a-i, b-iii, c-ii, d-iv

ugcnetcse-dec2012-paper3 operating-system match-the-following

Answer key

35.0.7 UGC NET CSE | June 2013 | Part 3 | Question: 59

A job has four pages A, B, C, D and the main memory has two page frames only. The job needs to process its pages in following order:

ABACABDBACD

Assuming that the page interrupt occurs when a new page is brought in the main memory, irrespective of whether the page is swapped out or not. The number of page interrupts in FIFO and LRU page replacement algorithms are

A. 9 and 7
B. 7 and 6
C. 9 and 8
D. 8 and 6

ugcnetcse-june2013-paper3 operating-system paging

Answer key

35.0.8 UGC NET CSE | September 2013 | Part 3 | Question: 23

Match the following :

(a)	Dangling pointer	(i)	Buffer replacement policy
(b)	Page fault	(ii)	Variable-length records
(c)	List representation	(iii)	Object identifier
(d)	Toss-immediate	(iv)	Pointer-swizzling

Codes :

A. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
B. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
C. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
D. (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

ugcnetcse-sep2013-paper3 operating-system page-fault

Answer key

35.0.9 UGC NET CSE | December 2014 | Part 2 | Question: 39

For the implementation of a paging scheme, suppose the average process size be x bytes, the page size be y bytes, and each page entry requires z bytes. The optimum page size that minimizes the total overhead due to the page table and the internal fragmentation loss is given by

A. $\frac{x}{2}$
B. $\frac{xz}{2}$
C. $\sqrt{2xz}$
D. $\frac{\sqrt{xz}}{2}$

ugcnetcse-dec2014-paper2 operating-system paging

Answer key

35.0.10 UGC NET CSE | December 2014 | Part 2 | Question: 40

In a demand paging memory system, page table is held in registers. The time taken to service a page fault is 8 m.sec. if an empty frame is available or if the replaced page is not modified, and it takes 20 m.secs., if the replaced page is modified. What is the average access time to service a page fault assuming that the page to be replaced is modified 70% of the time ?

A. 11.6 m.sec.
B. 16.4 m.sec.
C. 28 m.sec.
D. 14 m.sec.

ugcnetcse-dec2014-paper2 operating-system demand-paging

Answer key

35.0.11 UGC NET CSE | December 2013 | Part 2 | Question: 47

The virtual address generated by a CPU is 32 bits. The Translation Look-aside Buffer (TLB) can hold total 64 page table entries and a 4-way set associative (i.e. with 4-cache lines in the set). The page size is 4 KB. The minimum size of TLB tag is

A. 12 bits
B. 15 bits
C. 16 bits
D. 20 bits

ugcnetcse-dec2013-paper2 operating-system translation-lookaside-buffer

Answer key

35.0.12 UGC NET CSE | Junet 2015 | Part 3 | Question: 49

Match the following for operating system techniques with the most appropriate advantage :

	List-I		List-II
(a)	Spooling	(i)	Allows several jobs to improve CPU utilization
(b)	Multiprogramming	(ii)	Access to shared resources among geographically dispersed computers in a transparant way
(c)	Time sharing	(iii)	Overlapping I/O and computations
(d)	Distributed computing	(iv)	Allows many users to share a computer simultaneously by switching processor frequently

Codes :

A. (a)-(iii), (b)-(i), (c)-(ii), (d)-(iv)
B. (a)-(iii), (b)-(i), (c)-(iv), (d)-(ii)
C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
D. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

ugcnetcse-june2015-paper3 operating-system

Answer key

35.0.13 UGC NET CSE | Junet 2015 | Part 3 | Question: 52

Dining Philosopher's problem is a

A. Producer - consumer problem
B. Classical IPC problem
C. Starvation problem
D. Synchronization primitive

ugcnetcse-june2015-paper3 operating-system

Answer key

35.0.14 UGC NET CSE | December 2015 | Part 2 | Question: 27

Suppose that the number of instructions executed between page fault is directly proportional to the number of page frames allocated to a program. If the available memory is doubled, the mean interval between the page faults is also doubled. Further, consider that a normal instruction takes one microsecond, but if a page fault occurs, it takes 2001 microseconds. If a program takes 60 sec to run, during which time it gets 15,000 page faults, how long would it take to run twice as much memory were available?

A. 60 sec
B. 30 sec
C. 45 sec
D. 10 sec

ugcnetcse-dec2015-paper2 operating-system page-fault

Answer key

35.0.15 UGC NET CSE | December 2015 | Part 2 | Question: 29

A system has 4 processes and 5 allocatable resources. The current allocation and maximum needs are as follows:

	Allocated	**Maximum**	**Available**
Process **A**	1 0 2 1 1	1 1 2 1 3	0 0 x 1 1
Process **B**	2 0 1 1 0	2 2 2 1 0	
Process **C**	1 1 0 1 0	2 1 3 1 0	
Process **D**	1 1 1 1 0	1 1 2 2 1	

The smallest value of x for which the above system in safe state is

A. 1
B. 3
C. 2
D. 0

ugcnetcse-dec2015-paper2 operating-system

Answer key

35.0.16 UGC NET CSE | December 2015 | Part 3 | Question: 39

Consider a system with twelve magnetic tape drives and three processes P_1, P_2 and P_3. process P_1 requires maximum ten tape drives, process P_2 may need as many as four tape drives and P_3 may need upto nine tape drives. Suppose that at time t_1, process P_1 is holding five tape drives, process P_2 is holding two tape drives and process P_3 is holding three tape drives, At time t_1, system is in:

A. safe state
B. unsafe state
C. deadlocked state
D. starvation state

ugcnetcse-dec2015-paper3 operating-system

Answer key

35.0.17 UGC NET CSE | July 2016 | Part 2 | Question: 37

Suppose there are 4 processes in execution with 12 instances of a Resource R in a system.

The maximum need of each process and current allocation are given below :

Process	Max Need	Current Allocation
P_1	8	3
P_2	9	4
P_3	5	2
P_4	3	1

With reference to current allocation, is system safe? If so, what is the safe sequence?

A. No
B. Yes, $P_1 P_2 P_3 P_4$
C. Yes, $P_4 P_3 P_1 P_2$
D. Yes, $P_2 P_1 P_3 P_4$

ugcnetcse-july2016-paper2 operating-system

Answer key

35.0.18 UGC NET CSE | December 2010 | Part 2 | Question: 19

Optical storage is a

A. High-speed direct access storage device.
B. Low-speed direct access storage device.
C. Medium-speed direct access storage device.
D. High-speed sequential access storage device.

ugcnetcse-dec2010-paper2 operating-system

Answer key

35.0.19 UGC NET CSE | December 2010 | Part 2 | Question: 39

Page making process from main memory to disk is called

A. Interruption B. Termination C. Swapping D. None of the above

ugcnetcse-dec2010-paper2 operating-system paging

35.0.20 UGC NET CSE | June 2010 | Part 2 | Question: 36

Match the following :

(a)	Disk scheduling	1.	Robin-round
(b)	Batch processing	2.	SCAN
(c)	Time sharing	3.	LIFO
(d)	Interrupt processing	4.	FIFO

Codes :

A. (a)-3, (b)-4, (c)-2, (d)-1
B. (a)-4, (b)-3, (c)-2, (d)-1
C. (a)-2, (b)-4, (c)-1, (d)-3
D. (a)-1, (b)-4, (c)-3, (d)-2

ugcnetcse-june2010-paper2 operating-system match-the-following easy

Answer key

35.0.21 UGC NET CSE | June 2010 | Part 2 | Question: 40

Remote Computing Service involves the use of time sharing and _______.

A. Multi-processing

B. Interactive processing

C. Batch processing

D. Real-time processing

ugcnetcse-june2010-paper2 operating-system

Answer key

35.0.22 UGC NET CSE | August 2016 | Part 3 | Question: 51

An operating system supports a paged virtual memory, using a central processor with a cycle time of one microsecond. It costs an additional one microsecond to access a page other than the current one. Pages have 1000 words, and the paging device is a drum that rotates at 3000 revolutions per minute and transfers one million words per second. Further, one percent of all instructions executed accessed a page other than the current page. The instruction that accessed another page, 80% accessed a page already in memory and when a new page was required, the replaced page was modified 50% of the time. What is the effective access time on this system, assuming that the system is running only one process and the processor is idle during drum transfers ?

A. 30 microseconds
B. 34 microseconds
C. 60 microseconds
D. 68 microseconds

ugcnetcse-aug2016-paper3 operating-system paging

Answer key

35.0.23 UGC NET CSE | August 2016 | Part 3 | Question: 52

Consider the following page reference string :

1, 2, 3, 4, 2, 1, 5, 6, 2, 1, 2, 3, 7, 6, 3, 2, 1, 2, 3, 6

Which of the following options, gives the correct number of page faults related to LRU, FIFO, and optimal page replacement algorithms respectively, assuming 05 page frames and all frames are initially empty ?

A. 10,14,8 B. 8,10,7 C. 7,10,8 D. 7,10,7

ugcnetcse-aug2016-paper3 operating-system page-fault

Answer key

35.0.24 UGC NET CSE | December 2009 | Part 2 | Question: 37

A page fault

(A) is an error specific page.
(B) is an access to the page not currently in memory.
(C) occur when a page program occur in a page memory.
(D) page used in the previous page reference

ugcnetcse-dec2009-paper2 operating-system page-fault

Answer key

35.0.25 UGC NET CSE | June 2009 | Part 2 | Question: 10

Assume N segments in memory and a page size of P bytes. The wastage on account of internal fragmentation is :

1. $NP/2$ bytes
2. $P/2$ bytes
3. $N/2$ bytes
4. NP bytes

ugcnetcse-june2009-paper2

Answer key

35.0.26 UGC NET CSE | June 2009 | Part 2 | Question: 38

Variable partition memory management technique with compaction results in :

A. Reduction of fragmentation
B. Minimal wastage
C. Segment sharing
D. None of the above

ugcnetcse-june2009-paper2

35.0.27 UGC NET CSE | November 2017 | Part 3 | Question: 52

Consider a system with five processes P_0 through P_4 and three resource types A, B and C. Resource type A has seven instances, resource type B has two instances and resource type C has six instances suppose at time T_0 we have the following allocation.

Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P_0	0	1	0	0	0	0	0	0	0
P_1	2	0	0	2	0	2			
P_2	3	0	3	0	0	0			
P_3	2	1	1	1	0	0			
P_4	0	2	2	0	0	2			

Process	Allocation			Request			Available		
	A	B	C	A	B	C	A	B	C
P0	0	1	0	0	0	0	0	0	0
P1	2	0	0	2	0	2			
P2	3	0	3	0	0	0			
P3	2	1	1	1	0	0			
P4	0	2	2	0	0	2			

If we implement Deadlock detection algorithm we claim that system is ________

A. Semaphore
B. Deadlock state
C. Circular wait
D. Not in deadlock state

ugcnetcse-nov2017-paper3

Answer key

35.0.28 UGC NET CSE | July 2018 | Part 2 | Question: 58

Consider a virtual page reference string 1, 2, 3, 2, 4, 2, 5, 2, 3, 4. Suppose LRU page replacement algorithm is implemented with 3 page frames in main memory. Then the number of page faults are _____

A. 5
B. 7
C. 9
D. 10

ugcnetcse-july2018-paper2 operating-system page-fault paging

Answer key

35.0.29 UGC NET CSE | July 2018 | Part 2 | Question: 57

Page information memory is also called as Page Table. The essential contents in each entry of a page table is/are ____

A. Page Access information
B. Virtual Page number
C. Page Frame number
D. Both virtual page number and Page frame number

ugcnetcse-july2018-paper2 operating-system paging

Answer key

35.0.30 UGC NET CSE | July 2018 | Part 2 | Question: 53

In a multi-user operating system, 30 requests are made to use a particular resource per hour, on an average. The probability that no requests are made in 40 minutes, when arrival pattern is a poisson distribution, is ______

A. e^{-15}
B. $1-e^{-15}$
C. $1-e^{-20}$
D. e^{-20}

ugcnetcse-july2018-paper2 operating-system

Answer key

35.0.31 UGC NET CSE | July 2018 | Part 2 | Question: 52

In a paged memory, the page hit ratio is 0.40. The time required to access a page in secondary memory is equal to 120 ns. The time required to access a page in primary memory is 15 ns. The average time required to access a page is ____

A. 105
B. 68
C. 75
D. 78

ugcnetcse-july2018-paper2 operating-system paging

Answer key

35.0.32 UGC NET CSE | December 2018 | Part 2 | Question: 74

Suppose P, Q and R are co-operating processes satisfying Mutual Exclusion condition. Then, if the process Q is executing in its critical section then

A. Both 'P' and 'R' execute in critical section
B. Neither 'P' nor 'R' executes in their critical section
C. 'P' executes in critical section
D. 'R' executes in critical section

ugcnetcse-dec2018-paper2 operating-system

Answer key

35.0.33 UGC NET CSE | January 2017 | Part 2 | Question: 40

Distributed operating systems consist of:

A. Loosely coupled O.S. software on a loosely coupled hardware.
B. Loosely coupled O.S. software on a tightly coupled hardware.
C. Tightly coupled O.S. software on a loosely coupled hardware.
D. Tightly coupled O.S. software on a tightly coupled hardware.

ugcnetjan2017ii operating-system

Answer key

35.0.34 UGC NET CSE | January 2017 | Part 2 | Question: 37

In a paging system, it takes 30 ns to search translation Look-a-side Buffer (TLB) and 90 ns to access the main memory. If the TLB hit ratio is 70%, the effective memory access time is :

A. 48 ns B. 147ns C. 120ns D. 84ns

ugcnetjan2017ii operating-system paging

Answer key

35.0.35 UGC NET CSE | January 2017 | Part 3 | Question: 66

Names of some of the Operating Systems are given below:

i. MS-DOS
ii. XENIX
iii. OS/2

In the above list, following operating systems didn't provide multiuser facility.

A. (i) only
B. (i) and (ii)only
C. (ii) and (iii) only
D. (i),(ii) and (iii)

ugcnetcse-jan2017-paper3 operating-system

Answer key

35.0.36 UGC NET CSE | December 2004 | Part 2 | Question: 40

Match the following

(a) Disk scheduling	(1) Round robin
(b) Batch processing	(2) Scan
(c) Time sharing	(3) LIFO
(d) Interrupt processing	(4) FIFO

A. $a-3, b-4, c-2, d-1$
B. $a-4, b-3, c-2, d-1$
C. $a-2, b-4, c-1, d-3$
D. $a-3, b-4, c-1, d-2$

ugcnetcse-dec2004-paper2 operating-system match-the-following

Answer key

35.0.37 UGC NET CSE | December 2004 | Part 2 | Question: 38

Remote computing system involves the use of timesharing systems and :

A. Real time processing
B. Batch processing
C. Multiprocessing
D. All of the above

ugcnetcse-dec2004-paper2

Answer key

35.0.38 UGC NET CSE | December 2004 | Part 2 | Question: 37

In which of the following storage replacement strategies, is a program placed in the largest available hole in the memory ?

A. Best fit B. First fit C. Worst fit D. Buddy

ugcnetcse-dec2004-paper2

Answer key

35.0.39 UGC NET CSE | December 2004 | Part 2 | Question: 36

Semaphores are used to :

A. Synchronise critical resources to prevent deadlock
B. Synchronise critical resources to prevent contention
C. Do I/o
D. Facilitate memory management

ugcnetcse-dec2004-paper2

Answer key

35.0.40 UGC NET CSE | December 2006 | Part 2 | Question: 40

Loading operating system from secondary memory to primary memory is called ____________ .

A. Compiling B. Booting C. Refreshing D. Reassembling

ugcnetcse-dec2006-paper2 operating-system

Answer key

35.0.41 UGC NET CSE | December 2006 | Part 2 | Question: 36

An operating system is :

A. Collection of hardware components
B. Collection of input-output devices
C. Collection of software routines
D. All the above

ugcnetcse-dec2006-paper2 operating-system

Answer key

35.0.42 UGC NET CSE | December 2007 | Part 2 | Question: 40

The aging algorithm with a = 0.5 is used to predict run times. The previous four runs from oldest to most recent are 40, 20, 20, and 15 msec. The prediction for the next time will be :

A. 15 msec. B. 25 msec. C. 39 msec. D. 40 msec.

ugcnetcse-dec2007-paper2

Answer key

35.0.43 UGC NET CSE | December 2007 | Part 2 | Question: 38

Average process size = s bytes. Each page entry requires e bytes. The optimum page size is given by :

A. $\sqrt{(se)}$ B. $\sqrt{(2se)}$ C. s D. e

ugcnetcse-dec2007-paper2

Answer key

35.0.44 UGC NET CSE | December 2007 | Part 2 | Question: 37

A program has five virtual pages, numbered from 0 to 4. If the pages are referenced in the order 012301401234, with three page frames, the total number of page faults with FIFO will be equal to :

A. 0 B. 4 C. 6 D. 9

ugcnetcse-dec2007-paper2

Answer key

35.0.45 UGC NET CSE | December 2007 | Part 2 | Question: 36

How many states can a process be in ?

A. 2 B. 3 C. 4 D. 5

ugcnetcse-dec2007-paper2

Answer key

35.0.46 UGC NET CSE | December 2007 | Part 2 | Question: 25

The performance of a file system depends upon the cache hit rate. If it takes 1 msec to satisfy a request from the cache but 10 msec to satisfy a request if a disk read is needed, then the mean time (ms) required for a hit rate $'h'$ is given by :

A. 1 B. $h+10(1-h)$ C. $(1-h)+10h$ D. 10

ugcnetcse-dec2007-paper2

Answer key

35.0.47 UGC NET CSE | October 2020 | Part 2 | Question: 18

Consider a single-level page table system, with the page table stored in the memory. If the hit rate to TLB is 80%, and it takes 15 nanoseconds to search the TLB, and 150 nanoseconds to access the main memory, then what is the effective memory access time, in nanoseconds?

A. 185 B. 195 C. 205 D. 175

ugcnetcse-oct2020-paper2 operating-system translation-lookaside-buffer

Answer key

35.0.48 UGC NET CSE | October 2022 | Part 1 | Question: 59

Match List I with List II :

List I	List II
(A) Least frequently used	(I) Memory is distributed among processors
(B) Critical Section	(II) Page replacement policy in cache memory
(C) Loosely coupled multiprocessor system	(III) Program section that once begin must complete execution before another processor access the same shared resource
(D) Distributed operating system organization	(IV) O/S routines are distributed among

Choose the correct answer from the options given below :

A. (A)-(III), (B)-(II), (C)-(IV). (D)-(I)
B. (A)-(I), (B)-(II), (C)-(III), (D)-(IV)
C. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)
D. (A)-(II), (B)-(I), (C)-(Iii), (D)-(IV)

ugcnetcse-oct2022-paper1 operating-system

Answer key

35.0.49 UGC NET CSE

Given as 4 GB (4.3X10^9 bytes) of virtual space and typical page size of 4KB and each page table entry is 5 bytes. How many virtual pages would this imply? What is the size of the whole page table?

OPTIONS:

1. 107500 and 20480 bytes

2. 215000 and 40960 bytes

3. 10750 and 10240 bytes

4. 43000 and 1024 bytes

operating-system

35.1 Applications In Windows (1)

35.1.1 Applications In Windows: UGC NET CSE | December 2012 | Part 3 | Question: 35

In Win32, which function is used to create Windows Applications?

A. Win APP B. Win API C. Win Main D. Win Void

ugcnetcse-dec2012-paper3 applications-in-windows operating-system non-gatecse

Answer key

35.2 Bankers Algorithm (3)

35.2.1 Bankers Algorithm: UGC NET CSE | December 2005 | Part 2 | Question: 40

Banker's algorithm is used for __________ purpose :

A. Deadlock avoidance
B. Deadlock removal
C. Deadlock prevention
D. Deadlock continuations

ugcnetcse-dec2005-paper2 deadlock-prevention-avoidance-detection bankers-algorithm

Answer key

35.2.2 Bankers Algorithm: UGC NET CSE | June 2005 | Part 2 | Question: 38

Banker's algorithm is for

A. Dead lock Prevention
B. Dead lock Avoidance
C. Dead lock Detection
D. Dead lock creation

ugcnetcse-june2005-paper2 operating-system bankers-algorithm

Answer key

35.2.3 Bankers Algorithm: UGC NET CSE | June 2013 | Part 3 | Question: 61

An operating system using banker's algorithm for deadlock avoidance has ten dedicated devices (of same type) and has three processes P1, P2 and P3 with maximum resource requirements of 4, 5 and 8 respectively. There are two states of allocation of devices as follows:

State 1	Processes	P1	P2	P3
	Devices allocated	2	3	4
State 2	Processes	P1	P2	P3
	Devices allocated	0	2	4

Which of the following is correct?

A. State 1 is unsafe and State 2 is safe
B. State 1 is safe and State 2 is unsafe
C. Both State 1 and State 2 are safe
D. Both State 1 and State 2 are unsafe

ugcnetcse-june2013-paper3 operating-system bankers-algorithm deadlock-prevention-avoidance-detection

Answer key

35.3 Binary Semaphore (1)

35.3.1 Binary Semaphore: UGC NET CSE | June 2010 | Part 2 | Question: 39

In order to allow only one process to enter its critical section, binary semaphore are initialized to

A. 0 B. 1 C. 2 D. 3

ugcnetcse-june2010-paper2 operating-system binary-semaphore

Answer key

35.4 Counting Semaphores (1)

35.4.1 Counting Semaphores: UGC NET CSE | June 2019 | Part 2 | Question: 46

At a particular time of computation, the value of a counting semaphore is 7. Then 20 P (wait) operations and 15 V (signal) operations are completed on this semaphore. What is the resulting value of the semaphore?

A. 28 B. 12 C. 2 D. 42

ugcnetcse-june2019-paper2 counting-semaphores

Answer key

35.5 Critical Section (1)

35.5.1 Critical Section: UGC NET CSE | December 2014 | Part 2 | Question: 38

Which of the following conditions does not hold good for a solution to a critical section problem ?

A. No assumptions may be made about speeds or the number of CPU's.
B. No two processes may be simultaneously inside their critical sections.
C. Processes running outside its critical section may block other processes.
D. Processes do not wait forever to enter its critical section.

ugcnetcse-dec2014-paper2 operating-system critical-section

Answer key

35.6 Deadlock Prevention Avoidance Detection (9)

35.6.1 Deadlock Prevention Avoidance Detection: UGC NET CSE | August 2016 | Part 2 | Question: 36

Consider a system with seven processes A through G and six resources R through W.

Resource ownership is as follows :

process A holds R and wants T

process B holds nothing but wants T

process C holds nothing but wants S

process D holds $U$and wants S & T

process E holds T and wants V

process F holds W and wants S

process G holds V and wants U

Is the system deadlocked ? If yes, _______ processes are deadlocked.

A. No B. Yes, A, B, C C. Yes, D, E, G D. Yes, A, B, F

ugcnetcse-aug2016-paper2 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.2 Deadlock Prevention Avoidance Detection: UGC NET CSE | August 2016 | Part 2 | Question: 38

Consider a system having 'm' resources of the same type. These resources are shared by three processes P_1, P_2 and P_3 which have peak demands of $2, 5$ and 7 resources respectively. For what value of $'m'$ deadlock will not occur ?

A. 70 B. 14 C. 13 D. 7

ugcnetcse-aug2016-paper2 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.3 Deadlock Prevention Avoidance Detection: UGC NET CSE | August 2016 | Part 3 | Question: 50

Consider a system which have 'n' number of processes and 'm' number of resource types. The time complexity of the safety algorithm, which checks whether a system is in safe state or not, is of the order of :

A. $O(mn)$ B. $O(m^2n^2)$ C. $O(m^2n)$ D. $O(mn^2)$

ugcnetcse-aug2016-paper3 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.4 Deadlock Prevention Avoidance Detection: UGC NET CSE | December 2010 | Part 2 | Question: 40

A Dead-lock in an Operating System is

A. Desirable process
B. Undesirable process
C. Definite waiting process
D. All of the above

ugcnetcse-dec2010-paper2 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.5 Deadlock Prevention Avoidance Detection: UGC NET CSE | December 2011 | Part 2 | Question: 26

Dijikstra banking algorithm in an operating system, solves the problem of

A. Deadlock avoidance
B. Deadlock recovery
C. Mutual exclusion
D. Context switching

ugcnetcse-dec2011-paper2 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.6 Deadlock Prevention Avoidance Detection: UGC NET CSE | December 2013 | Part 3 | Question: 71

Simplest way of deadlock recovery is

A. Roll back
B. Preempt resource
C. Lock one of the processes
D. Kill one of the processes

ugcnetcse-dec2013-paper3 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.7 Deadlock Prevention Avoidance Detection: UGC NET CSE | December 2014 | Part 3 | Question: 52

An operating system has 13 tape drives. There are three processes $P1, P2$ & $P3$. Maximum requirement of $P1$ is 11 tape drives, $P2$ is 5 tape drives and $P3$ is 8 tape drives. Currently, $P1$ is allocated 6 tape drives, $P2$ is allocated 3 tape drives and $P3$ is allocated 2 tape drives. Which of the following sequences represent a safe state?

1. $P2P1P3$
2. $P2P3P1$
3. $P1P2P3$
4. $P1P3P2$

ugcnetcse-dec2014-paper3 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.8 Deadlock Prevention Avoidance Detection: UGC NET CSE | December 2018 | Part 2 | Question: 70

Suppose a system has 12 instances of some resource with n processes competing for that resource. Each process may require 4 instances of the resources. The maximum value of n for which the system never enters into deadlock is

A. 3 B. 4 C. 5 D. 6

ugcnetcse-dec2018-paper2 operating-system deadlock-prevention-avoidance-detection

Answer key

35.6.9 Deadlock Prevention Avoidance Detection: UGC NET CSE | June 2019 | Part 2 | Question: 48

A computer has six tape drives with n processes competing for them. Each process may need two drives. What is the maximum value of n for the system to be deadlock free?

A. 5 B. 4 C. 3 D. 6

ugcnetcse-june2019-paper2 deadlock-prevention-avoidance-detection

Answer key

35.7 Delay (1)

35.7.1 Delay: UGC NET CSE | July 2016 | Part 3 | Question: 53

A Multicomputer with 256 CPUs is organized as 16×16 grid . What is the worst case delay(in hops) that a message might have to take

A. 16 B. 15 C. 32 D. 30

ugcnetcse-july2016-paper3 operating-system delay

Answer key

35.8 Dijkstras Bankers Algorithm (1)

35.8.1 Dijkstras Bankers Algorithm: UGC NET CSE | September 2013 | Part 3 | Question: 69

A system contains 10 units of resource of same type. The resource requirement and current allocation of these resources for three processes P, Q and R are as follows :

	P	Q	R
Maximum requirement	8	7	5
Current allocation	4	1	3

Now, consider the following resource requests:

i. P makes a request for 2 resource units
ii. Q makes request for 2 resource units
iii. R makes a request for 2 resource units

For a safe state, which of the following options must be satisfied?

A. Only request i
B. Only request ii
C. Only request iii
D. Request i and ii

ugcnetcse-sep2013-paper3 operating-system dijkstras-bankers-algorithm

Answer key

35.9 Directory Structure (1)

35.9.1 Directory Structure: UGC NET CSE | December 2014 | Part 2 | Question: 23

The directory can be viewed as _______ that translates filenames into their directory entries.

A. Symbol table B. Partition C. Swap space D. Cache

ugcnetcse-dec2014-paper2 operating-system file-system directory-structure

Answer key

35.10 Disk (10)

35.10.1 Disk: UGC NET CSE | December 2013 | Part 2 | Question: 49

How much space will be required to store the bit map of a 1.3 GB disk with 512 bytes block size?

A. 332.8 KB
B. 83.6 KB
C. 266.2 KB
D. 256.6 KB

operating-system disk ugcnetcse-dec2013-paper2

Answer key

35.10.2 Disk: UGC NET CSE | December 2015 | Part 2 | Question: 28

Consider a disk with 16384 bytes per track having a rotation time of 16 msec and average seek time of 40 msec. What is the time in msec to read a block of 1024 bytes from this disk?

A. 57 msec B. 49 msec C. 48 msec D. 17 msec

ugcnetcse-dec2015-paper2 operating-system disk

Answer key

35.10.3 Disk: UGC NET CSE | December 2018 | Part 2 | Question: 18

Consider a disk pack with 32 surfaces, 64 tracks and 512 sectors per pack. 256 bytes of data are stored in a bit serial manner in a sector. The number of bits required to specify a particular sector in the disk is

A. 18 B. 19 C. 20 D. 22

ugcnetcse-dec2018-paper2 operating-system disk

Answer key

35.10.4 Disk: UGC NET CSE | June 2013 | Part 3 | Question: 64, UGCNET-Dec2015-II: 49

A UNIX file system has 1 KB block size and 4-byte disk addresses. What is the maximum file size if the inode contains ten direct block entries, one single indirect block entry, one double indirect block entry and one triple indirect block entry?

A. 30 GB B. 64 GB C. 16 GB D. 1 GB

ugcnetcse-june2013-paper3 ugcnetcse-dec2015-paper2 operating-system disk

Answer key

35.10.5 Disk: UGC NET CSE | October 2020 | Part 2 | Question: 17

Suppose you have a Linux file system where the block size is $2K$ bytes, a disk address is 32 bits, and an i–node contains the disk addresses of the first 12 direct blocks of file, a single indirect block and a double indirect block. Approximately, what is the largest file that can be represented by an i–node?

A. 513 Kbytes B. 513 MBytes C. 537 Mbytes D. 537 KBytes

ugcnetcse-oct2020-paper2 operating-system disk

Answer key

35.10.6 Disk: UGC NET CSE | October 2020 | Part 2 | Question: 91

The question given below: concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.

If T is the capacity of a track in bytes, and S is the capacity of each surface in bytes, then $(T, S) =$ _______

A. $(50K, 50000K)$
B. $(25K, 25000K)$
C. $(25K, 50000K)$
D. $(40K, 36000K)$

ugcnetcse-oct2020-paper2 operating-system disk

Answer key ☟

35.10.7 Disk: UGC NET CSE | October 2020 | Part 2 | Question: 92

The question given below: concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.

What is the capacity of the disk, in bytes?

A. $25,000K$
B. $500,000K$
C. $250,000K$
D. $50,000K$

ugcnetcse-oct2020-paper2 operating-system disk

Answer key ☟

35.10.8 Disk: UGC NET CSE | October 2020 | Part 2 | Question: 93

The question given below: concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.

Given below are two statements:

Statement I: The disk has a total number of 2000 cylinders.

Statement II: 51200 bytes is not a valid block size for the disk.

In the light of the above statements, choose the correct answer from the options given below:

A. Both Statement I and Statement II are true
B. Both Statement I and Statement II are false
C. Statement I is correct but Statement II is false
D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 operating-system disk

Answer key ☟

35.10.9 Disk: UGC NET CSE | October 2020 | Part 2 | Question: 94

The question given below concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.

If the disk platters rotate at 5400 rpm (revolutions per minute), then approximately what is the maximum rotational delay?

A. 0.011 seconds
B. 0.11 seconds
C. 0.0011 seconds
D. 1.1 seconds

ugcnetcse-oct2020-paper2 operating-system disk

Answer key ☟

35.10.10 Disk: UGC NET CSE | October 2020 | Part 2 | Question: 95

The question given below concern a disk with a sector size of 512 bytes, 2000 tracks per surface, 50 sectors per track, five double-sided platters, and average seek time of 10 milliseconds.

If one track of data can be transferred per revolution, then what is the data transfer rate?

A. $2,850$ KBytes/second
B. $4,500$ KBytes/second

C. $5,700$ KBytes/second

D. $2,250$ KBytes/second

ugcnetcse-oct2020-paper2 operating-system disk

Answer key

35.11 Disk Cpu Memory Interrupt (1)

35.11.1 Disk Cpu Memory Interrupt: UGC NET CSE | June 2019 | Part 2 | Question: 42

Match List-I with List-II:

	List-I		List-II
(a)	Disk	(i)	Thread
(b)	CPU	(ii)	Signal
(c)	Memory	(iii)	File system
(d)	Interrupt	(iv)	Virtual address space

Choose the correct option from those given below:

A. a-i; b-ii; c-iii; d-iv

B. a-iii; b-i; c-iv; d-ii

C. a-ii; b-i; c-iv; d-iii

D. a-ii; b-iv; c-iii; d-i

ugcnetcse-june2019-paper2 disk-cpu-memory-interrupt

Answer key

35.12 Disk Scheduling (9)

35.12.1 Disk Scheduling: UGC NET CSE | December 2012 | Part 2 | Question: 5 UGCNET-June2013-II: 47

If the disk head is located initially at 32, find the number of disk moves required with FCFS if the disk queue of I/O blocks requests are 98. 37, 14, 124, 65, 67.

A. 239

B. 310

C. 321

D. 325

ugcnetcse-dec2012-paper2 operating-system disk-scheduling ugcnetcse-june2013-paper2

Answer key

35.12.2 Disk Scheduling: UGC NET CSE | December 2013 | Part 2 | Question: 48

Consider a disk queue with request for input/output to block cylinders 98, 183, 37, 122, 14, 124, 65, 67 in that order. Assume that disk head is initially positioned at cylinder 53 and moving towards cylinder number 0. The total number of head movements using Shortest Seek Time First (SSTF) and SCAN algorithms are respectively

A. 236 and 252 cylinders

B. 640 and 236 cylinders

C. 235 and 640 cylinders

D. 235 and 252 cylinders

ugcnetcse-dec2013-paper2 operating-system disk-scheduling

Answer key

35.12.3 Disk Scheduling: UGC NET CSE | December 2014 | Part 3 | Question: 51

Consider an imaginary disk with 40 cylinders. A request come to read a block on cylinder 11. While the seek to cylinder 11 is in progress, new requests come in for cylinders $1, 36, 16, 34, 9$ and 12 in that order. The number of arm motions using shortest seek first algorithm is

A. 111

B. 112

C. 60

D. 61

ugcnetcse-dec2014-paper3 operating-system disk-scheduling

Answer key

35.12.4 Disk Scheduling: UGC NET CSE | January 2017 | Part 3 | Question: 50

Consider a disk queue with I/O requests on the following cylinders in their arriving order:

$6, 10, 12, 54, 97, 73, 128, 15, 44, 110, 34, 45$. The disk head is assumed to be at cylinder 23 and moving in the direction of decreasing number of cylinders. Total number of cylinders in the disk is 150. The disk head movement using SCAN – scheduling algorithm is:

A. 172
B. 173
C. 227
D. 228

ugcnetcse-jan2017-paper3 operating-system disk-scheduling

Answer key

35.12.5 Disk Scheduling: UGC NET CSE | July 2016 | Part 2 | Question: 38

If the Disk head is located initially at track 32, find the number of disk moves required with FCFS scheduling criteria if the disk queue of I/O blocks requests are 98, 37, 14, 124, 65, 67

A. 320
B. 322
C. 321
D. 319

ugcnetcse-july2016-paper2 operating-system disk-scheduling

Answer key

35.12.6 Disk Scheduling: UGC NET CSE | June 2012 | Part 3 | Question: 74

On a disk with 1000 cylinders (0 to 999) find the number of tracks, the disk arm must move to satisfy all the requests in the disk queue. Assume the last request service was at track 345 and the head is moving toward track 0. The queue in FIFO order contains requests for the following tracks:

123, 874, 692, 475, 105, 376

(Assume scan algorithm)

A. 2013
B. 1219
C. 1967
D. 1507

ugcnetcse-june2012-paper3 operating-system disk-scheduling

Answer key

35.12.7 Disk Scheduling: UGC NET CSE | Junet 2015 | Part 2 | Question: 36

A disk drive has 100 cylinders, numbered 0 to 99. Disk requests come to the disk driver for cylinders $12, 26, 24, 4, 42, 8$ and 50 in that order. The driver is currently serving a request at a cylinder 24. A seek takes 6 msec per cylinder moved. How much seek time is needed for shortest seek time first (SSTF) algorithm?

A. 0.984 sec
B. 0.396 sec
C. 0.738 sec
D. 0.42 sec

ugcnetcse-june2015-paper2 operating-system disk-scheduling

Answer key

35.12.8 Disk Scheduling: UGC NET CSE | October 2020 | Part 2 | Question: 16

Consider a disk system having 60 cylinders. Disk requests are received by a disk drive for cylinders $10, 22, 20, 2, 40, 6$ and 38, in that order. Assuming the disk head is currently at cylinder 20, what is the time taken to satisfy all the requests if it takes 2 milliseconds to move from one cylinder to adjacent one and Shortest Seek Time First (SSTF) algorithm is used?

A. 240 milliseconds
B. 96 milliseconds
C. 120 milliseconds
D. 112 milliseconds

ugcnetcse-oct2020-paper2 operating-system disk-scheduling

Answer key

35.12.9 Disk Scheduling: UGC NET CSE | September 2013 | Part 2 | Question: 46

Consider the input/output (I/O) request made at different instants of time directed at a hypothetical disk having 200 tracks as given in the following table :

Serial No.	1	2	3	4	5
TranckNo.	12	85	40	100	75
Time of Arrival	65	80	110	100	175

Assume that :

Current head position is at track no. 65

Direction of last movement is towards higher number of tracks.

Current clock time is 160 milliseconds

Head movement time per track is 1 millisecond

"look" is a variant of "SCAN" disk-arm scheduling algorithm. In this algorithm, if no more I/O requests are left in current direction, the disk head reveres its direction. The seek time in Shortest Seek First (SSF) and "look" disk-arm scheduling algorithms respectively are :

A. 144 and 123 milliseconds
B. 143 and 123 milliseconds
C. 149 and 124 milliseconds
D. 256 and 186 milliseconds

ugcnetsep2013ii operating-system disk-scheduling

Answer key

35.13 Distributed Computing (1)

35.13.1 Distributed Computing: UGC NET CSE | December 2012 | Part 3 | Question: 2

The efficiency (E) and speed up (sp) for Multiprocessor with p processors satisfies:

A. $E \leq p$ and $s_p \leq p$
B. $E \leq 1$ and $s_p \leq p$
C. $E \leq p$ and $s_p \leq 1$
D. $E \leq 1$ and $s_p \leq 1$

ugcnetcse-dec2012-paper3 operating-system distributed-computing

Answer key

35.14 Distributed System (1)

35.14.1 Distributed System: UGC NET CSE | December 2008 | Part 2 | Question: 24

An example of distributed OS is:

A. Amoeba
B. UNIX
C. MS-DOS
D. MULTICS

ugcnetcse-dec2008-paper2 operating-system distributed-system

Answer key

35.15 Dynamic Linking (1)

35.15.1 Dynamic Linking: UGC NET CSE | July 2016 | Part 2 | Question: 34

Which of the following is not typically a benefit of dynamic linking?

I. Reduction in overall program execution time
II. Reduction in overall space consumption in memory
III. Reduction in overall space consumption on disk
IV. Reduction in the cost of software updates

A. I and IV
B. I only
C. II and III
D. IV only

ugcnetcse-july2016-paper2 operating-system dynamic-linking

Answer key

35.16 Effective Memory Access (1)

35.16.1 Effective Memory Access: UGC NET CSE | June 2009 | Part 2 | Question: 19

Suppose it takes 100 ns to access a page table and 20 ns to access associative memory with a 90% hit rate, the average access time equals :

A. 28
B. 20
C. 90
D. 100

ugcnetcse-june2009-paper2 operating-system effective-memory-access

Answer key

35.17 File System (7)

35.17.1 File System: UGC NET CSE | August 2016 | Part 3 | Question: 53

Consider a file currently consisting of 50 blocks. Assume that the file control block and the index block is already in memory. If a block is added at the end (and the block information to be added is stored in memory), then how many disk I/O operations are required for indexed (single-level) allocation strategy ?

A. 1
B. 101
C. 27
D. 0

ugcnetcse-aug2016-paper3 operating-system file-system

Answer key

35.17.2 File System: UGC NET CSE | December 2013 | Part 3 | Question: 72

The directory structure used in Unix file system is called

A. Hierarchical directory
B. Tree structured directory
C. Directed acrylic graph
D. Graph structured directory

ugcnetcse-dec2013-paper3 operating-system file-system

Answer key

35.17.3 File System: UGC NET CSE | December 2014 | Part 3 | Question: 50

How many disk blocks are required to keep list of free disk blocks in a 16 GB hard disk with 1 kB block size using linked list of free disk blocks ? Assume that the disk block number is stored in 32 bits.

A. 1024 blocks
B. 16794 blocks
C. 20000 blocks
D. 1048576 blocks

ugcnetcse-dec2014-paper3 operating-system file-system

Answer key

35.17.4 File System: UGC NET CSE | June 2012 | Part 2 | Question: 10

Which command is the fastest among the following?

A. COPY TO <NEW FILE>
B. COPY STRUCTURE TO <NEW FILE>
C. COPY FILE <FILE 1> <FILE 2>
D. COPY TO MFILE-DAT DELIMITED

ugcnetcse-june2012-paper2 operating-system file-system

Answer key

35.17.5 File System: UGC NET CSE | June 2014 | Part 2 | Question: 34

Match the following :

	List – I		**List – II**
a.	Contiguous allocation	i.	This scheme supports very large file sizes.
b.	Linked allocation	ii.	This allocation technique supports only sequential files.
c.	Indexed allocation	iii.	The number of disks required to access file is minimal.
d.	Multi-level indexed	iv.	This technique suffers from maximum wastage of space in storing pointers.

Codes :

A. a-iii; b-iv; c-ii; d-i
B. a-iii; b-ii; c-iv; d-i
C. a-i; b-ii; c-iv; d-iii
D. a-i; b-iv; c-ii; d-iii

ugcnetcse-june2014-paper2 operating-system file-system

Answer key ☟

35.17.6 File System: UGC NET CSE | Junet 2015 | Part 3 | Question: 12

In the indexed scheme of blocks to a file, the maximum possible size of the file depends on:

A. The number of blocks used for index and the size of index
B. Size of Blocks and size of Address
C. Size of index
D. Size of block

ugcnetcse-june2015-paper3 operating-system file-system

Answer key ☟

35.17.7 File System: UGC NET CSE | Junet 2015 | Part 3 | Question: 53

In _____ allocation method for disk block allocation in a file system, insertion and deletion of blocks in a file is easy

A. Index
B. Linked
C. Contiguous
D. Bit Map

ugcnetcse-june2015-paper3 operating-system file-system

Answer key ☟

35.18 Fragmentation (1)

35.18.1 Fragmentation: UGC NET CSE | July 2018 | Part 2 | Question: 56

Which of the following statements are true?

i. External Fragmentation exists when there is enough total memory space to satisfy a request but the available space is contiguous.
ii. Memory Fragmentation can be internal as well as external
iii. One solution to external Fragmentation is compaction

A. i and ii only
B. i and iii only
C. ii and iii only
D. i, ii and iii

ugcnetcse-july2018-paper2 operating-system fragmentation

Answer key ☟

35.19 Hard Disk Drive (1)

35.19.1 Hard Disk Drive: UGC NET CSE | December 2011 | Part 2 | Question: 29

The maximum amount of information that is available in one portion of the disk access arm for a removal disk pack (without further movement of the arm with multiple heads)

A. A plate of data
B. A cylinder of data
C. A track of data
D. A block of data

ugcnetcse-dec2011-paper2 operating-system system hard-disk-drive

35.20 Hit Ratio (1)

35.20.1 Hit Ratio: UGC NET CSE | September 2013 | Part 2 | Question: 45

The hit ratio of a Transaction Look Aside Buffer (TLAB) is 80%. It takes 20 nanoseconds (ns) to search TLAB and 100 ns to access main memory. The effective memory access time is _____

A. 36 ns
B. 140 ns
C. 122 ns
D. 40 ns

ugcnetsep2013ii operating-system cache-memory hit-ratio

Answer key ☟

35.21 IO Handling (1)

35.21.1 IO Handling: UGC NET CSE | July 2018 | Part 2 | Question: 54

Normally user programs are prevented from handling I/O directly by I/O instructions in them. For CPUs having explicit I/O instructions, such I/O protection is ensured by having the I/O instructions privileged. In a CPU with memory mapped I/O, there is no explicit I/O instruction. Which one of the following is true for a CPU with memory mapped I/O?

A. I/O protection is ensured by operating system routines
B. I/O protection is ensured by a hardware trap
C. I/O protection is ensured during system configuration
D. I/O protection is not possible

ugcnetcse-july2018-paper2 operating-system io-handling

Answer key ☟

35.22 Input Output (2)

35.22.1 Input Output: UGC NET CSE | January 2017 | Part 2 | Question: 38

Match the following w.r.t Input/Output management :

	List – I		List – II
a.	Device controller	i.	Extracts information from the controller register and store it in data buffer
b.	Device driver	ii.	I/O scheduling
c.	Interrupt handler	iii.	Performs data transfer
d.	Kernel I/O subsystem	iv.	Processing of I/O request

	a	b	c	d
A.	iii	iv	i	ii
B.	ii	i	iv	iii
C.	iv	i	ii	iii
D.	i	iii	iv	ii

ugcnetjan2017ii operating-system input-output

Answer key ☟

35.22.2 Input Output: UGC NET CSE | October 2020 | Part 2 | Question: 65

Match List I with List II

	List I		List II
(A)	Handshaking	(I)	I/O interface informs the CPU that device is ready for the transfer
(B)	Programmed I/O	(II)	requires two control signals working in opposite directions
(C)	Interrupt-initiated I/O	(III)	has local memory & control large set of I/O devices
(D)	I/O processor	(IV)	require CPU to check the I/O flag & perform transfer

Choose the correct answer from the options given below:

A. $A-I, B-II, C-III, D-IV$
B. $A-II, B-IV, C-III, D-I$
C. $A-II, B-IV, C-I, D-III$
D. $A-IV, B-III, C-II, D-I$

ugcnetcse-oct2020-paper2 operating-system input-output

Answer key ☟

35.23 Interrupts (2)

35.23.1 Interrupts: UGC NET CSE | December 2011 | Part 2 | Question: 28

On receiving an interrupt from an I/O device, the CPU

A. Halts for predetermined time.
B. Branches off to the interrupt service routine after completion of the current instruction.
C. Branches off to the interrupt service routine immediately.
D. Hands over control of address bus and data bus to the interrupting device.

ugcnetcse-dec2011-paper2 operating-system interrupts

Answer key ☟

35.23.2 Interrupts: UGC NET CSE | December 2014 | Part 3 | Question: 02

For switching from a CPU user mode to the supervisor mode following type of interrupt is most appropriate

A. Internal interrupts
B. External interrupts
C. Software interrupts
D. None of the above

ugcnetcse-dec2014-paper3 operating-system interrupts

Answer key ☟

35.24 Java (1)

35.24.1 Java: UGC NET CSE | June 2012 | Part 3 | Question: 13

Which of the following is scheme to deal with deadlock?

A. Time out
B. Time in
C. Both A and B
D. None of the above

ugcnetcse-june2012-paper3 operating-system java deadlock-prevention-avoidance-detection

Answer key ☟

35.25 Linux (1)

35.25.1 Linux: UGC NET CSE | July 2018 | Part 2 | Question: 55

Which UNIX/Linux command is used to make all files and sub-directions in the directory "progs" executable by all users?

A. chmod - R a + x progs
B. chmod - R 222 progs
C. chmod - X a + x progs
D. chmod - X 222 progs

ugcnetcse-july2018-paper2 unix linux operating-system non-gatecse

Answer key ☟

35.26 Memory Management (8)

35.26.1 Memory Management: UGC NET CSE | December 2008 | Part 2 | Question: 40

Which of the following OS treats hardware as a file system?

A. $UNIX$
B. DOS
C. $Windows\ NT$
D. None of the above

ugcnetcse-dec2008-paper2 operating-system memory-management

Answer key ☟

35.26.2 Memory Management: UGC NET CSE | December 2009 | Part 2 | Question: 39

A program is located in the smallest available hole in the memory is _________

(A) best – fit

(B) first – bit

(C) worst – fit

(D) buddy

ugcnetcse-dec2009-paper2 operating-system memory-management

Answer key ☟

35.26.3 Memory Management: UGC NET CSE | December 2011 | Part 2 | Question: 30

Consider a logical address space of 8 pages of 1024 words mapped with memory of 32 frames. How many bits are there in the physical address ?

1. 9 bits
2. 11 bits
3. 13 bits
4. 15 bits

ugcnetcse-dec2011-paper2 operating-system memory-management

Answer key ☟

35.26.4 Memory Management: UGC NET CSE | December 2012 | Part 2 | Question: 14

Given a memory partitions of 100 K, 500 K, 200 K, 300 K and 600 K (in order) and processes of 212 K, 417 K, 112 K and 426 K (in order), using the first fit algorithm, in which partition would the process requiring 426 K be placed?

A. 500 K
B. 200 K
C. 300 K
D. 600 K

ugcnetcse-dec2012-paper2 operating-system memory-management

Answer key

35.26.5 Memory Management: UGC NET CSE | December 2012 | Part 2 | Question: 30

Which of the following memory allocation scheme suffers from external fragmentation?

A. Segmentation
B. Pure demand paging
C. Swapping
D. Paging

ugcnetcse-dec2012-paper2 operating-system memory-management

Answer key

35.26.6 Memory Management: UGC NET CSE | December 2015 | Part 3 | Question: 37

Match the following with respect to various memory management algorithms :

	List-I		List-II
(a)	Demand paging	(i)	degree of multiprogramming
(b)	Segmentation	(ii)	working set
(c)	Dynamic partitions	(iii)	supports user view of memory
(d)	Fixed partitions	(iv)	compaction

Codes :

A. (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)
B. (a)-(ii), (b)-(iii), (c)-(i), (d)-(iv)
C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
D. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)

operating-system memory-management ugcnetcse-dec2015-paper3

Answer key

35.26.7 Memory Management: UGC NET CSE | January 2017 | Part 3 | Question: 49

A memory management system has 64 pages with 512 bytes page size. Physical memory consists of 32 page frames. Number of bits required in logical and physical address are respectively:

A. 14 and 15
B. 14 and 29
C. 15 and 14
D. 16 and 32

ugcnetcse-jan2017-paper3 operating-system memory-management

Answer key

35.26.8 Memory Management: UGC NET CSE | Junet 2015 | Part 3 | Question: 51

What is the most appropriate function of Memory Management Unit (MMU)?

A. It is an associative memory to store TLB
B. It is a technique of supporting multiprogramming by creating dynamic partitions
C. It is a chip to map virtual address to physical address
D. It is an algorithm to allocate and deallocate main memory to a process

ugcnetcse-june2015-paper3 operating-system memory-management

Answer key

35.27 Mmu (1)

35.27.1 Mmu: UGC NET CSE | December 2015 | Part 3 | Question: 38

Function of memory management unit is :

A. Address translation
B. Memory Allocation

C. Cache management

D. All of the above

ugcnetcse-dec2015-paper3 operating-system mmu

Answer key

35.28 Monitors (1)

35.28.1 Monitors: UGC NET CSE | December 2014 | Part 3 | Question: 53

Monitor is an Interprocess Communication (IPC) technique which can be described as

A. It is higher level synchronization primitive and is a collection of procedures, variables, and data structures grouped together in a special package.
B. It is a non-negative integer which apart from initialization can be acted upon by wait and signal operations.
C. It uses two primitives, send and receive which are system calls rather than language constructs.
D. It consists of the IPC primitives implemented as system calls to block the process when they are not allowed to enter critical region to save CPU time.

ugcnetcse-dec2014-paper3 operating-system monitors

Answer key

35.29 Multiprogramming Os (1)

35.29.1 Multiprogramming Os: UGC NET CSE | December 2012 | Part 3 | Question: 7

Which of the following features will characterize an OS as multi-programmed OS?

i. More than one program loaded into main memory at the same time.
ii. If a program waits for certain event another program is immediately scheduled
iii. If the execution of a program terminates, another program is immediately scheduled

A. i only

B. i and ii only

C. i and iii only

D. i, ii, and iii only

ugcnetcse-dec2012-paper3 operating-system multiprogramming-os

Answer key

35.30 Mutual Exclusion (1)

35.30.1 Mutual Exclusion: UGC NET CSE | June 2012 | Part 3 | Question: 65

Consider the methods used by processes P_1 and P_2 for accessing their critical sections. The initial values of shared Boolean variables S_1 and S_2 are randomly assigned,

P_1	P_2
while (S_1==S_2);	while (S_1==S2);
critical section	critical section
S_1=S_2;	S_1=S_2;

Which one of the following statements describes the properties achieved?

A. Mutual exclusion but not progress

B. Progress but not mutual exclusion

C. Neither mutual exclusion nor progress

D. Both mutual exclusion and progress

ugcnetcse-june2012-paper3 operating-system mutual-exclusion critical-section

Answer key

35.31 Non Gatecse (3)

35.31.1 Non Gatecse: UGC NET CSE | December 2012 | Part 3 | Question: 40

Which of the following operating system is better for implementing client-server network?

A. Windows 95

B. Windows 98

C. Windows 2000

D. All of these

ugcnetcse-dec2012-paper3 operating-system computer-networks non-gatecse

Answer key

35.31.2 Non Gatecse: UGC NET CSE | December 2013 | Part 3 | Question: 69

To place a sound into a word document, following feature of windows is used

A. Clip board B. Task switching C. C Win App D. OLE

ugcnetcse-dec2013-paper3 operating-system non-gatecse

Answer key

35.31.3 Non Gatecse: UGC NET CSE | October 2022 | Part 1 | Question: 1

For multiprocessor system, interconnection network - cross bar switch is an example of

A. Non blocking network
B. Blocking network
C. That varies from connection to connection
D. Recurrent network

ugcnetcse-oct2022-paper1 non-gatecse

Answer key

35.32 Page Replacement (9)

35.32.1 Page Replacement: UGC NET CSE | August 2016 | Part 2 | Question: 37

Suppose that the virtual Address space has eight pages and physical memory with four page frames. If LRU page replacement algorithm is used, ______ number of page faults occur with the reference string.

0 2 1 3 5 4 6 3 7 4 7 3 3 5 5 3 1 1 1 7 2 3 4 1

A. 11 B. 12 C. 10 D. 9

ugcnetcse-aug2016-paper2 operating-system least-recently-used page-replacement

Answer key

35.32.2 Page Replacement: UGC NET CSE | December 2018 | Part 2 | Question: 72

Suppose for a process P, reference to pages in order are $1, 2, 4, 5, 2, 1, 2, 4$. Assume that main memory can accomodate 3 pages and the main memory has already 1 and 2 in the order 1 – first, 2 – second. At this moment, assume FIFO Page Replacement Algorithm is used then the number of page faults that occur to complete the execution of process P is

A. 4 B. 3 C. 5 D. 6

ugcnetcse-dec2018-paper2 operating-system page-replacement page-fault

Answer key

35.32.3 Page Replacement: UGC NET CSE | July 2016 | Part 2 | Question: 36

Consider the reference string:

0 1 2 3 0 1 4 0 1 2 3 4

If FIFO page replacement algorithm is used, then the number of page faults with three page frames and four page frames are ____ and ___ respectively.

A. 10, 9 B. 9, 9 C. 10, 10 D. 9, 10

ugcnetcse-july2016-paper2 operating-system page-replacement page-fault

Answer key

35.32.4 Page Replacement: UGC NET CSE | June 2012 | Part 2 | Question: 33

Consider the following page trace:

4,3,2,1,4,3,5,4,3,2,1,5

Percentage of page fault that would occur if FIFO page replacement algorithm is used with number of frames for the JOB m=4 will be

A. 8
B. 9
C. 10
D. 12

operating-system page-replacement ugcnetcse-june2012-paper2

Answer key

35.32.5 Page Replacement: UGC NET CSE | June 2014 | Part 3 | Question: 35

Consider a program that consists of 8 pages (from 0 to 7) and we have 4 page frames in the physical memory for the pages. The page reference string is :

1 2 3 2 5 6 3 4 6 3 7 3 1 5 3 6 3 4 2 4 3 4 5 1

The number of page faults in LRU and optimal page replacement algorithms are respectively (without including initial page faults to fill available page frames with pages) :

A. 9 and 6
B. 10 and 7
C. 9 and 7
D. 10 and 6

ugcnetjune2014iii operating-system page-replacement

Answer key

35.32.6 Page Replacement: UGC NET CSE | June 2019 | Part 2 | Question: 43

Consider that a process has been allocated 3 frames and has a sequence of page referencing as $1, 2, 1, 3, 7, 4, 5, 6, 3, 1$. What shall be the difference in page faults for the above string using the algorithms of LRU and optimal page replacement for referencing the string?

A. 2
B. 0
C. 1
D. 3

ugcnetcse-june2019-paper2 page-replacement least-recently-used

Answer key

35.32.7 Page Replacement: UGC NET CSE | Junet 2015 | Part 2 | Question: 38

A LRU page replacement is used with 4 page frames and eight pages. How many page faults will occur with the reference string 0172327103 if the four frames are initially empty?

A. 6
B. 7
C. 5
D. 8

ugcnetcse-june2015-paper2 operating-system page-replacement

Answer key

35.32.8 Page Replacement: UGC NET CSE | October 2020 | Part 2 | Question: 15

Consider a hypothetical machine with 3 pages of physical memory, 5 pages of virtual memory, and $< A, B, C, D, A, B, E, A, B, C, D, E, B, A, B >$ as the stream of page reference by an application. If P and Q are the number of page faults that the application would incur with FIFO and LRU page replacement algorithms respectively, then $(P, Q) =$ _______(Assuming enough space for storing 3 page frames)

A. $(11,10)$
B. $(12,11)$
C. $(10,11)$
D. $(11,12)$

ugcnetcse-oct2020-paper2 operating-system page-replacement

Answer key

35.32.9 Page Replacement: UGC NET CSE | September 2013 | Part 3 | Question: 71

Consider a main memory with three page frames for the following page reference string:

5, 4, 3, 2, 1, 4, 3, 5, 4, 3, 4, 1, 4

Assuming that the execution of process is initialized after loading page 5 in memory,, the number of page faults in FIFO and second chance replacement respectively are

A. 8 and 9 B. 10 and 11 C. 7 and 9 D. 9 and 8

ugcnetcse-sep2013-paper3 operating-system page-replacement

Answer key

35.33 Pipelining (1)

35.33.1 Pipelining: UGC NET CSE | September 2013 | Part 2 | Question: 50

The speed up of a pipeline processing over an equivalent non-pipeline processing is defined by the ration:

where $n \rightarrow$ no. of tasks

$t_n \rightarrow$ time of completion of each tasks

$k \rightarrow$ no. of segments of pipeline

$t_p \rightarrow$ clock cycle time

$S \rightarrow$ speed up ratio

A. $S = \frac{nt_n}{(k+n-1)t_p}$

B. $S = \frac{nt_n}{(k+n+1)t_p}$

C. $S = \frac{nt_n}{(k-n+1)t_p}$

D. $S = \frac{(k+n-1)t_p}{nt_n}$

ugcnetsep2013ii operating-system pipelining

Answer key

35.34 Process (2)

35.34.1 Process: UGC NET CSE | December 2014 | Part 2 | Question: 37

A specific editor has $200K$ of program text, $15K$ of initial stack, $50K$ of initialized data, and $70K$ of bootstrap code. If five editors are started simultaneously, how much physical memory is needed if shared text is used?

A. $1135K$ B. $335K$ C. $1065K$ D. $320K$

ugcnetcse-dec2014-paper2 operating-system process

Answer key

35.34.2 Process: UGC NET CSE | June 2013 | Part 3 | Question: 63

Working set model is used in memory management to implement the concept of

A. Swapping

B. Principal of Locality

C. Segmentation

D. Thrashing

ugcnetcse-june2013-paper3 operating-system process

Answer key

35.35 Process Model (2)

35.35.1 Process Model: UGC NET CSE | December 2008 | Part 2 | Question: 21

N processes are waiting for I/O. A process spends a fraction p of its time in I/O wait state. The CPU utilization is given by :

1. 1-p-N
2. 1-pN
3. pN
4. p-N

ugcnetcse-dec2008-paper2 operating-system process-model

Answer key

35.35.2 Process Model: UGC NET CSE | December 2008 | Part 2 | Question: 22

If holes are half as large as processes, the fraction of memory wasted in holes is :

A. $2/3$
B. $1/2$
C. $1/3$
D. $1/5$

ugcnetcse-dec2008-paper2 operating-system process-model

Answer key

35.36 Process Scheduling (25)

35.36.1 Process Scheduling: UGC NET CSE | August 2016 | Part 2 | Question: 39

Five jobs A, B, C, D and E are waiting in Ready Queue. Their expected runtimes are $9, 6, 3, 5$ and x respectively. All jobs entered in Ready queue at time zero. They must run in _____ order to minimize average response time if $3 < x < 5$.

A. B, A, D, E, C
B. C, E, D, B, A
C. E, D, C, B, A
D. C, B, A, E, D

ugcnetcse-aug2016-paper2 operating-system process-scheduling

Answer key

35.36.2 Process Scheduling: UGC NET CSE | August 2016 | Part 2 | Question: 40

Consider three CPU intensive processes $P1, P2, P3$ which require $20, 10$ and 30 units of time, arrive at times $1,$ and 7 respectively. Suppose operating system is implementing Shortest Remaining Time first (preemptive scheduling) algorithm, then _____ context switches are required (suppose context switch at the beginning of Ready queue and at the end of Ready queue are not counted).

A. 3
B. 2
C. 4
D. 5

ugcnetcse-aug2016-paper2 operating-system process-scheduling

Answer key

35.36.3 Process Scheduling: UGC NET CSE | December 2006 | Part 2 | Question: 37

____________ is one of pre-emptive scheduling algorithm.

A. Shortest-Job-first
B. Round-robin
C. Priority based
D. Shortest-Job-next

ugcnetcse-dec2006-paper2 operating-system process-scheduling

Answer key

35.36.4 Process Scheduling: UGC NET CSE | December 2006 | Part 2 | Question: 38

A software to create a Job Queue is called ____________ .

A. Linkage editor
B. Interpreter
C. Driver
D. Spooler

ugcnetcse-dec2006-paper2 operating-system process-scheduling

Answer key

35.36.5 Process Scheduling: UGC NET CSE | December 2009 | Part 2 | Question: 36

In the process management Round-robin method is essentially the pre-emptive version of _________

(A) FILO

(B) FIFO

(C) SSF

(D) Longest time first

ugcnetcse-dec2009-paper2 operating-system process-scheduling

Answer key

35.36.6 Process Scheduling: UGC NET CSE | December 2012 | Part 2 | Question: 20

The problem of indefinite blockage of low-priority jobs in general priority scheduling algorithm can be solved using:

A. Parity bit B. Aging C. Compaction D. Timer

ugcnetcse-dec2012-paper2 operating-system process-scheduling

Answer key

35.36.7 Process Scheduling: UGC NET CSE | December 2013 | Part 2 | Question: 46

Consider a preemptive priority based scheduling algorithm based on dynamically changing priority. Larger priority number implies higher priority. When the process is waiting for CPU in the ready queue (but not yet started execution), its priority changes at a rate a=2. When it starts running, its priority changes at a rate b=1. All the processes are assigned priority value 0 when they enter ready queue. Assume that the following processes want to execute:

Process ID	Arrival Time	Service Time
P1	0	4
P2	1	1
P3	2	2
P4	3	1

The time quantum q=1. When two processes want to join ready queue simultaneously, the process which has not executed recently is given priority. The finish time of processes P1, P2, P3 and P4 will respectively be

A. 4, 5, 7 and 8 B. 8, 2, 7 and 5 C. 2, 5, 7 and 8 D. 8, 2, 5 and 7

ugcnetcse-dec2013-paper2 operating-system process-scheduling

Answer key

35.36.8 Process Scheduling: UGC NET CSE | December 2014 | Part 2 | Question: 36

Consider the following justifications for commonly using the two-level CPU scheduling:

I. It is used when memory is too small to hold all the ready processes.

II. Because its performance is same as that of the $FIFO$.

III. Because it facilitates putting some set of processes into memory and a choice is made from that.

IV. Because it does not allow to adjust the set of in-core processes.

Which of the following is true ?

A. I, III and IV B. I and II C. III and IV D. I and III

ugcnetcse-dec2014-paper2 operating-system process-scheduling

Answer key

35.36.9 Process Scheduling: UGC NET CSE | December 2018 | Part 2 | Question: 73

Consider the following set of processes and the length of CPU burst time given in milliseconds :

Process	CPU Burst time (ms)
P_1	5
P_2	7
P_3	6
P_4	4

Assume that processes being scheduled with Round-Robin Scheduling Algorithm with time quantum 4 ms. Then the waiting time for P_4 is ______ ms

A. 0
B. 4
C. 12
D. 6

ugcnetcse-dec2018-paper2 operating-system process-scheduling round-robin-scheduling

Answer key ☟

35.36.10 Process Scheduling: UGC NET CSE | January 2017 | Part 3 | Question: 52

Some of the criteria for calculation of priority of a process are:

i. Processor utilization by an individual process.
ii. Weight assigned to a user or group of users
iii. Processor utilization by a user or group of processes

In fair scheduler, priority is calculated based on:

A. only (i) and (ii)
B. only (i) and (iii)
C. (i) ,(ii) and (iii)
D. only (ii) and (iii)

ugcnetcse-jan2017-paper3 operating-system process-scheduling

Answer key ☟

35.36.11 Process Scheduling: UGC NET CSE | July 2016 | Part 2 | Question: 40

A scheduling Algorithm assigns priority proportional to the waiting time of a process. Every process starts with priority zero (lowest priority). The scheduler reevaluates the process priority for every 'T' time units and declares next process to be scheduled. If teh process have no I/O operations and all arrive at time zero, then the schduler implements ___ criteria

A. Priority scheduling
B. Rond Robin Scheduling
C. Shortest Job First
D. FCFS

ugcnetcse-july2016-paper2 operating-system process-scheduling

Answer key ☟

35.36.12 Process Scheduling: UGC NET CSE | July 2018 | Part 2 | Question: 60

In which of the following scheduling criteria, context switching will never take place?

A. ROUND ROBIN
B. Preemptive SJF
C. Non-preemptive SJF
D. Preemptive priority

ugcnetcse-july2018-paper2 process-scheduling operating-system

Answer key ☟

35.36.13 Process Scheduling: UGC NET CSE | June 2005 | Part 2 | Question: 39

Which is the correct definition of a valid process transition in an operating system ?

A. Wake up : Ready $\rightarrow$ Running
B. Dispatch: Ready $\rightarrow$ Running
C. Block : Ready $\rightarrow$ Running
D. Timer run out: Ready $\rightarrow$ Blocked

ugcnetcse-june2005-paper2 operating-system process-scheduling

Answer key

35.36.14 Process Scheduling: UGC NET CSE | June 2010 | Part 2 | Question: 38

_______ is one of pre-emptive scheduling algorithm.

A. RR
B. SSN
C. SSF
D. Priority based

ugcnetcse-june2010-paper2 operating-system process-scheduling

Answer key

35.36.15 Process Scheduling: UGC NET CSE | June 2012 | Part 2 | Question: 29

Pre-emptive scheduling is the strategy of temporarily suspending a running process

A. before the CPU time slice expires
B. to allow starving processes to run
C. when it requires I/O
D. to avoid collision

operating-system process-scheduling ugcnetcse-june2012-paper2

Answer key

35.36.16 Process Scheduling: UGC NET CSE | June 2012 | Part 2 | Question: 30

In round robin CPU scheduling as time quantum is increased the average turn around time

A. increases
B. decreases
C. remains constant
D. varies irregularly

ugcnetcse-june2012-paper2 operating-system process-scheduling

Answer key

35.36.17 Process Scheduling: UGC NET CSE | June 2013 | Part 3 | Question: 58

Consider the following processes with time slice of 4 milliseconds (I/O requests are ignored):

Process	A	B	C	D
Arrival time	0	1	2	3
CPU Cycle	8	4	9	5

The average turnaround time of these processes will be

A. 19.25 milliseconds
B. 18.25 milliseconds
C. 19.5 milliseconds
D. 18.5 milliseconds

ugcnetcse-june2013-paper3 operating-system process-scheduling

Answer key

35.36.18 Process Scheduling: UGC NET CSE | June 2014 | Part 2 | Question: 32

Match the following :

	List – I		**List – II**
a.	Multilevel feedback queue	i.	Time-slicing
b.	FCFS	ii.	Criteria to move processes between queues
c.	Shortest Process next	iii.	Batch Processing
d.	Round robin scheduling	iv.	Exponential smoothening

Codes :

A. a-i; b-iii; c-ii; d-iv
B. a-iv; b-iii; c-ii; d-i

C. a-iii; b-i; c-iv; d-i

D. a-ii; b-iii; c-iv; d-i

ugcnetcse-june2014-paper2 operating-system process-scheduling

Answer key

35.36.19 Process Scheduling: UGC NET CSE | June 2014 | Part 3 | Question: 34

Consider a uniprocessor system where new processes arrive at an average of five processes per minute and each process needs an average of 6 seconds of service time. What will be the CPU utilization ?

A. $80\,\%$

B. $50\,\%$

C. $60\,\%$

D. $30\,\%$

ugcnetjune2014iii operating-system process-scheduling

Answer key

35.36.20 Process Scheduling: UGC NET CSE | June 2014 | Part 3 | Question: 36

Which of the following statements is not true about disk-arm scheduling algorithms ?

A. SSTF (shortest seek time first) algorithm increases performance of FCFS.
B. The number of requests for disk service are not influenced by file allocation method.
C. Caching the directories and index blocks in main memory can also help in reducing disk arm movements.
D. SCAN and C-SCAN algorithms are less likely to have a starvation problem.

ugcnetjune2014iii operating-system process-scheduling

Answer key

35.36.21 Process Scheduling: UGC NET CSE | June 2019 | Part 2 | Question: 45

Consider three CPU intensive processes, which require 10, 20 and 30 units of time and arrive at times 0, 2 and 6 respectively. How many context switches are needed if the operating system implements a shortest remaining time first scheduling algorithm? Do not count the context switches at time zero and at the end.

A. 4

B. 2

C. 3

D. 1

ugcnetcse-june2019-paper2 process-scheduling

Answer key

35.36.22 Process Scheduling: UGC NET CSE | June 2023 | Part 2: 22

Consider the following table of arrival time and burst time for three processes $P0,P1\ P2$:

Process	arrival time	Burst time
P0	0 ms	7
P1	1 ms	3
P2	2 ms	7

The pre-emptive shortest job first scheduling algorithm is used. Scheduling is carried out only at arrival or completion of a process. What is the average waiting time for the three processes?

A. 3 ms

B. 3.67 ms

C. 4.47 ms

D. 4 ms

ugcnetcse-june2023-paper2 operating-system process-scheduling

Answer key

35.36.23 Process Scheduling: UGC NET CSE | Junet 2015 | Part 3 | Question: 50

Which of the following statements is not true for Multi Level Feedback Queue processor scheduling algorithm?

A. Queues have different priorities

B. Each queue may have different scheduling algorithm
C. Processes are permanently assigned to a queue
D. This algorithm can be configured to match a specific system under design

ugcnetcse-june2015-paper3 operating-system process-scheduling

Answer key

35.36.24 Process Scheduling: UGC NET CSE | September 2013 | Part 2 | Question: 44

Match the following :

	List – I Process state transition		List – II Reason for transition
(a.)	Ready → Running	(i.)	Request made by the process is satisfied or an event for which it was waiting to occurs
(b.)	Blocked → Ready	(ii.)	Process wishes to wait for some action by another process
(c.)	Running → Blocked	(iii.)	The process is dispatched
(d.)	Running → Ready	(iv.)	The Process is preempted

Codes :

A. a-iii, b-i, c-ii, d-iv
B. a-iv, b-i, c-iii, d-ii
C. a-iv, b-iii, c-i, d-ii
D. a-iii, b-iv, c-ii, d-i

ugcnetsep2013ii operating-system process-scheduling match-the-following easy

Answer key

35.36.25 Process Scheduling: UGC NET CSE | September 2013 | Part 3 | Question: 70

Consider the following set of processes with the length of CPU burst time in milliseconds (ms) :

Process	A	B	C	D	E
Burst time	6	1	2	1	5
Priority	3	1	3	4	2

Assume that processes are stored in ready queue in following order :

A – B – C – D – E

Using round robin scheduling with time slice of 1 ms, the average turn around time is

A. 8.4 ms
B. 12.4 ms
C. 9.2 ms
D. 9.4 ms

ugcnetcse-sep2013-paper3 operating-system process-scheduling

Answer key

35.37 Process Synchronization (2)

35.37.1 Process Synchronization: UGC NET CSE | December 2018 | Part 2 | Question: 69

To overcome difficulties in Readers-Writers problem, which of the following statement/s is/are ***true***?

i. Writers are given exclusive access to shared objects.
ii. Readers are given exclusive access to shared objects.
iii. Both Readers and Writers are given exclusive access to shared objects.

Choose the correct answer from the code given below :

Code :

A. (i) only
B. (ii) only
C. (iii) only
D. Both (ii) and (iii)

ugcnetcse-dec2018-paper2 operating-system process-synchronization

Answer key

35.37.2 Process Synchronization: UGC NET CSE | July 2018 | Part 2 | Question: 51

At a particular time of computation, the value of a counting semaphore is 10. Then 12 P operations and "x" V operations were performed on this semaphore. If the final value of semaphore is 7, x will be

A. 8 B. 9 C. 10 D. 11

ugcnetcse-july2018-paper2 operating-system process-synchronization

Answer key

35.38 Ram (1)

35.38.1 Ram: UGC NET CSE | December 2008 | Part 2 | Question: 41

In which of the following, ready to execute processes must be present in RAM ?

A. multiprocessing
B. multiprogramming
C. multitasking
D. all of the above

ugcnetcse-dec2008-paper2 operating-system ram

Answer key

35.39 Resource Allocation (2)

35.39.1 Resource Allocation: UGC NET CSE | June 2012 | Part 2 | Question: 31

Resources are allocated to the process on non-sharable basis is

A. mutual exclusion B. hold and wait
C. no pre-emption D. circular wait

ugcnetcse-june2012-paper2 operating-system resource-allocation

Answer key

35.39.2 Resource Allocation: UGC NET CSE | June 2014 | Part 2 | Question: 33

Consider a system with five processes P_0 through P_4 and three resource types R_1, R_2 and R_3 . Resource type R_1 has 10 instances, R_2 has 5 instances and R_3 has 7 instances. Suppose that at time T_0, the following snapshot of the system has been taken :

Allocation

	R_1	R_2	R_3
P_0	0	1	0
P_1	2	0	0
P_2	3	0	2
P_3	2	1	1
P_4	0	2	2

Max

R_1	R_2	R_3
7	5	3
3	2	2
9	0	2
2	2	2

4	3	3

Available

R_1	R_2	R_3
3	3	2

Assume that now the process P_1 requests one additional instance of type R_1 and two instances of resource type R_3. The state resulting after this allocation will be

A. Ready state
B. Safe state
C. Blocked state
D. Unsafe state

ugcnetcse-june2014-paper2 operating-system resource-allocation

Answer key

35.40 Runtime Environment (1)

35.40.1 Runtime Environment: UGC NET CSE | December 2009 | Part 2 | Question: 31

In an absolute loading scheme which loader function is accomplished by assembler ?

(A) re-allocation

(B) allocation

(C) linking

(D) loading

ugcnetcse-dec2009-paper2 operating-system runtime-environment

Answer key

35.41 Scheduling (4)

35.41.1 Scheduling: UGC NET CSE | December 2008 | Part 2 | Question: 23

An example of a non pre-emptive scheduling algorithm is :

A. Round Robin
B. Priority Scheduling
C. Shortest job first
D. 2 level scheduling

ugcnetcse-dec2008-paper2 operating-system scheduling

35.41.2 Scheduling: UGC NET CSE | December 2010 | Part 2 | Question: 38

WINDOWS is a _________ operating.

A. Real time
B. Multi-user
C. Preemptive
D. Non-preemptive

ugcnetcse-dec2010-paper2 operating-system scheduling

Answer key

35.41.3 Scheduling: UGC NET CSE | January 2017 | Part 2 | Question: 39

Which of the following scheduling algorithms may cause starvation ?

a. First-come-first-served
b. Round Robin
c. Priority

d. Shortest process next

e. Shortest remaining time first

A. a, c and e
B. c, d and e
C. b, d and e
D. b, c and d

ugcnetjan2017ii operating-system scheduling

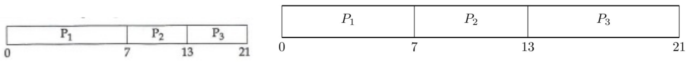
Answer key

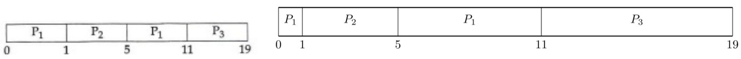
35.41.4 Scheduling: UGC NET CSE | July 2018 | Part 2 | Question: 59

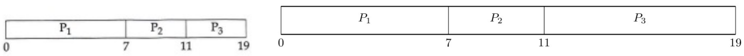
Consider the following three processes with the arrival time and CPU burst time given in miliseconds:

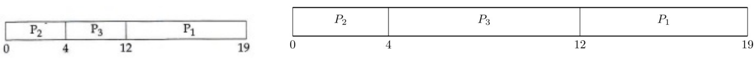
Process	Arrival Time	Burst Time
P_1	0	7
P_2	1	4
P_3	2	8

The Gantt Chart for preemptive SJF scheduling algorithm is _______

A. 

B. 

C. 

D. 

ugcnetcse-july2018-paper2 operating-system scheduling process-scheduling

Answer key

35.42 Seek Latency (1)

35.42.1 Seek Latency: UGC NET CSE | June 2019 | Part 2 | Question: 16

For a magnetic disk with concentric circular tracks, the seek latency is not linearly proportional to the seek distance due to

A. non-uniform distribution of requests
B. arm starting or stopping inertia
C. higher capacity of tracks on the periphery of the platter
D. use of uniform arm scheduling policies

ugcnetcse-june2019-paper2 seek-latency

Answer key

35.43 Semaphore (3)

35.43.1 Semaphore: UGC NET CSE | December 2009 | Part 2 | Question: 38

A semaphore count of negative n means (s = – n) that the queue contains ________ waiting processes.

(A) n + 1

(B) n

(C) n – 1

(D) 0

ugcnetcse-dec2009-paper2 operating-system semaphore

Answer key

35.43.2 Semaphore: UGC NET CSE | January 2017 | Part 2 | Question: 36

There are three processes P_1, P_2 and P_3 sharing a semaphore for synchronizing a variable. Initial value of semaphore is one. Assume that negative value of semaphore tells us how many processes are waiting in queue. Processes access the semaphore in following order :

a. $P2$ needs to access

b. $P1$ needs to access

c. $P3$ needs to access

d. $P2$ exits critical section

e. $P1$ exits critical section

The final value of semaphore will be :

A. 0 B. 1 C. -1 D. -2

ugcnetjan2017ii operating-system semaphore

Answer key

35.43.3 Semaphore: UGC NET CSE | June 2013 | Part 3 | Question: 60

Suppose S and Q are two semaphores initialized to 1. P1 and P2 are two processes which are sharing resources.

P1 has statements	**P2 has statements**
wait(S);	wait(Q);
wait(Q);	wait(S);
critical-section 1;	critical-section 2;
signal(S);	signal(Q);
Signal(Q);	signal(S);

Their execution may sometimes lead to an undesirable situation called

A. Starvation B. Race condition C. Multithreading D. Deadlock

ugcnetcse-june2013-paper3 operating-system semaphore

Answer key

35.44 Shell Script (7)

35.44.1 Shell Script: UGC NET CSE | July 2016 | Part 3 | Question: 68

Which of the following statements is not correct with reference to cron daemon in UNIX OS?

A. The cron daemon is the standard tool for running commands on a predetermined-schedule
B. It starts when the system boots and runs as long as the system is up
C. Cron reads configuration files that contain list of command lines and the times at which they invoked
D. Crontab for individual users are not stored

ugcnetcse-july2016-paper3 unix shell-script

Answer key

35.44.2 Shell Script: UGC NET CSE | July 2016 | Part 3 | Question: 69

In Unix, files can be protected by assigning each one a 9-bit mode called rights bits, now, consider the following two statements:

I. A mode of 641 (octal) means that the owner can read and write the file, other members of the owner's group can read it, and users can execute only
II. A mode of 100 (octal) allows the owner to execute the file , but prohibits all other access

Which of the following statements is correct with reference to above statements?

A. Only I is correct
B. Only II is correct
C. Both I and II are correct
D. Both I and II are incorrect

ugcnetcse-july2016-paper3 unix shell-script

Answer key ☟

35.44.3 Shell Script: UGC NET CSE | June 2019 | Part 2 | Question: 50

Which of the following UNIX/Linux pipes will count the number of lines in all the files having .c and .h as their extension in the current working directory?

A. cat $*\cdot$ ch | wc -1
B. cat $*\cdot$[c-h] | wc -1
C. cat $*\cdot$[ch] | ls -1
D. cat $*\cdot$[ch] | wc -1

ugcnetcse-june2019-paper2 shell-script

Answer key ☟

35.44.4 Shell Script: UGC NET CSE | Junet 2015 | Part 2 | Question: 39

What does the following command do? grep -vn "abc" x

A. It will print all of the lines in the file x that match the search string "abc"
B. It will print all of the lines in file x that do not match the search string "abc"
C. It will print the total number of lines in the file x that match the search string "abc"
D. It will print the specific line numbers of the file x in which there is a match for string "abc"

ugcnetcse-june2015-paper2 operating-system shell-script non-gatecse

Answer key ☟

35.44.5 Shell Script: UGC NET CSE | Junet 2015 | Part 3 | Question: 75

The unix command: $ vi file1 file2

A. Edits file1 and stores the contents of file1 in file2
B. Both files i.e., file1 and file2 can be edited using 'ex' command to travel between the files
C. Both files can be edited using 'mv' command to move between the files
D. Edits file1 first, saves it and then edits file2

ugcnetcse-june2015-paper3 shell-script

Answer key ☟

35.44.6 Shell Script: UGC NET CSE | September 2013 | Part 3 | Question: 72

Which of the following shell scripts will produce the output “my first script”?

A. for i in my first script { echo –i $i }
B. for my first script; do echo –n; done
C. for i in my first script; do echo –i $i; done
D. for n in my first script; do echo –i $i; done

ugcnetcse-sep2013-paper3 unix shell-script non-gatecse

Answer key ☟

35.44.7 Shell Script: UGC NET CSE | September 2013 | Part 3 | Question: 75

Which of the following statements is not true for UNIX Operating System?

A. Major strength of UNIX Operating System is its open standards which enables large number of organizations ranging from academicia to industries to participate in its development

B. UNIX kernel uses modules with well specified interfaces and provides advantages like simplified tesing and maintenance of kernel.
C. UNIX is kernel based operating system with two main components viz. process management subsystem and file management subsystem.
D. All devices are represented as files which simplify the management of I/O devices and files. The directories structure used is directed acyclic graph.

ugcnetcse-sep2013-paper3 unix shell-script

Answer key

35.45 Sjf (1)

35.45.1 Sjf: UGC NET CSE | December 2008 | Part 2 | Question: 8

Four jobs J1, J2, J3 and J4 are waiting to be run. Their expected run times are 9, 6, 3 and 5 respectively. In order to minimize average response time, the jobs should be run in the order:

(A) J1 J2 J3 J4 (B) J4 J3 J2 J1

(C) J3 J4 J1 J2 (D) J3 J4 J2 J1

ugcnetcse-dec2008-paper2 operating-system sjf scheduling

Answer key

35.46 Sstf (1)

35.46.1 Sstf: UGC NET CSE | June 2019 | Part 2 | Question: 41

Consider a disk system with 100 cylinders. The requests to access the cylinders occur in the following sequences:

$$4, 34, 10, 7, 19, 73, 2, 15, 6, 20$$

Assuming that the head is currently at cylinder 50, what is the time taken to satisfy all requests if it takes 1 ms to move from the cylinder to adjacent one and the shortest seek time first policy is used?

A. 357 ms B. 238 ms C. 276 ms D. 119 ms

ugcnetcse-june2019-paper2 disk-scheduling sstf

Answer key

35.47 Storage Devices (1)

35.47.1 Storage Devices: UGC NET CSE | December 2014 | Part 3 | Question: 49

Various storage devices used by an operating system can be arranged as follows in increasing order of accessing speed :

A. Magnetic tapes $\rightarrow$ magnetic disks $\rightarrow$ optical disks $\rightarrow$ electronic disks $\rightarrow$ main memory $\rightarrow$ cache $\rightarrow$ registers
B. Magnetic tapes $\rightarrow$ magnetic disks $\rightarrow$ electronic disks $\rightarrow$ optical disks $\rightarrow$ main memory $\rightarrow$ cache $\rightarrow$ registers
C. Magnetic tapes $\rightarrow$ electronic disks $\rightarrow$ magnetic disks $\rightarrow$ optical disks $\rightarrow$ main memory $\rightarrow$ cache $\rightarrow$ registers
D. Magnetic tapes $\rightarrow$ optical disks $\rightarrow$ magnetic disks $\rightarrow$ electronic disks $\rightarrow$ main memory $\rightarrow$ cache $\rightarrow$ registers

ugcnetcse-dec2014-paper3 storage-devices accessing-speed

Answer key

35.48 System Call (2)

35.48.1 System Call: UGC NET CSE | November 2017 | Part 3 | Question: 75

Match the following WINDOWS system calls and UNIX system calls with reference to process control and File manipulation.

	Windows		UNIX
(a)	Create – process ( )	(i)	Open ( )
(b)	WaitForSingleObject ( )	(ii)	Close ( )
(c)	CreateFile ( )	(iii)	Fork ( )
(d)	CloseHandle ( )	(iv)	Wait ( )

A. $(a) - (iii)$ $(b) - (iv)$ $(c) - (i)$ $d - (ii)$
B. $(a) - (iv)$ $(b) - (iii)$ $(c) - (i)$ $d - (ii)$
C. $(a) - (iv)$ $(b) - (iii)$ $(c) - (ii)$ $d - (i)$
D. $(a) - (iii)$ $(b) - (iv)$ $(c) - (ii)$ $d - (i)$

ugcnetcse-nov2017-paper3 operating-system system-call

Answer key

35.48.2 System Call: UGC NET CSE | October 2020 | Part 2 | Question: 84

Assuming that the system call fork() never fails, consider the following C program $P1$ and $P2$ executed on a UNIX/Linux system:

/* P1 */	/* P2 */

int main(){
fork();
fork();
fork();
printf("Happy \n");
}
	int main(){
	fork();
	printf("Happy \n");
	fork();
	printf("Happy \n");
	fork();
	printf("Happy \n");
	}

Statement I: $P1$ displays "Happy" 8 times

Statement II: $P2$ displays "Happy" 12 times

In the light of the above statements, choose the correct answer from the options given below

A. Both Statement I and Statement II are true
B. Both Statement I and Statement II are false
C. Statement I is correct but Statement II is false
D. Statement I is incorrect but Statement II is true

ugcnetcse-oct2020-paper2 operating-system system-call

Answer key

35.49 System Reliability (1)

35.49.1 System Reliability: UGC NET CSE | August 2016 | Part 3 | Question: 43

An Operating System (OS) crashes on the average once in 30 days, that is, the Mean Time Between Failures (MTBF) = 30 days. When this happens, it takes 10 minutes to recover the OS, that is, the Mean Time To Repair (MTTR) = 10 minutes. The availability of the OS with these reliability figures is approximately :

A. 96.97% B. 97.97% C. 99.009% D. 99.97%

ugcnetcse-aug2016-paper3 operating-system system-reliability

Answer key

35.50 Threads (4)

35.50.1 Threads: UGC NET CSE | December 2012 | Part 3 | Question: 32

A thread is a light weight process.

In the above statement, weight refers to

A. time
B. number of resources
C. speed
D. all of the above

ugcnetcse-dec2012-paper3 operating-system threads

Answer key

35.50.2 Threads: UGC NET CSE | December 2013 | Part 3 | Question: 75

Possible thread states in Windows 2000 operating system include:

A. Ready, running and waiting
B. Ready, standby, running, waiting, transition and terminated
C. Ready, running, waiting, transition and terminated
D. Standby, running, waiting, transition and terminated

ugcnetcse-dec2013-paper3 operating-system threads non-gatecse

Answer key

35.50.3 Threads: UGC NET CSE | January 2017 | Part 3 | Question: 53

One of the disadvantages of user level threads compared to Kernel level thread is

A. If a user level thread of a process executes a system call, all threads in that process are blocked.
B. Scheduling is application dependent.
C. Thread switching doesn't require kernel mode privileges.
D. The library procedures invoked for thread management in user level threads are local procedures.

ugcnetcse-jan2017-paper3 operating-system threads

Answer key

35.50.4 Threads: UGC NET CSE | June 2019 | Part 2 | Question: 44

Which of the following are NOT shared by the threads of the same process?

a. Stack
b. Registers
c. Address space
d. Message queue

A. a and d
B. b and c
C. a and b
D. a, b and c

ugcnetcse-june2019-paper2 threads

Answer key

35.51 Transaction and Concurrency (1)

35.51.1 Transaction and Concurrency: UGC NET CSE | Junet 2015 | Part 2 | Question: 37

Let $P-i$ and P_j two processes, R be the set of variables read from memory, and W be the set of variables written to memory. For the concurrent execution of two processes p_i and P_j which of the conditions are not true?

A. $R(P_i) \cap W(P_j) = \Phi$
B. $W(P_i) \cap R(P_j) = \Phi$
C. $R(P_i) \cap R(P_j) = \Phi$
D. $W(P_i) \cap W(P_j) = \Phi$

ugcnetcse-june2015-paper2 transaction-and-concurrency

Answer key

35.52 Unix (24)

35.52.1 Unix: UGC NET CSE | August 2016 | Part 3 | Question: 49

The Unix Operating System Kernel maintains two key data structures related to processes, the process table and the user structure. Now, consider the following two statements :

I. The process table is resident all the time and contain information needed for all processes, even those that are not currently in memory.

II. The user structure is swapped or paged out when its associated process is not in memory, in order not to waste memory on information that is not needed.

Which of the following options is correct with reference to above statements ?

1. Only (I) is correct.
2. Only (II) is correct.
3. Both (I) and (II) are correct.
4. Both (I) and (II) are wrong.

ugcnetcse-aug2016-paper3 operating-system unix

Answer key

35.52.2 Unix: UGC NET CSE | December 2009 | Part 2 | Question: 40

The Unix command used to find out the number of characters in a file is

(A) nc

(B) wc

(C) chcnt

(D) lc

ugcnetcse-dec2009-paper2 unix operating-system

Answer key

35.52.3 Unix: UGC NET CSE | December 2012 | Part 2 | Question: 36

In UNIX, which of the following command is used to set the task priority?

A. init B. nice C. kill D. PS

ugcnetcse-dec2012-paper2 unix operating-system

Answer key

35.52.4 Unix: UGC NET CSE | December 2012 | Part 3 | Question: 37

Everything below the system call interface and above the physical hardware is known as

A. Kernel B. Bus C. Shell D. Stub

ugcnetcse-dec2012-paper3 unix operating-system non-gatecse

Answer key

35.52.5 Unix: UGC NET CSE | December 2013 | Part 2 | Question: 50

Linux operating system uses

A. Affinity scheduling
B. Fair Preemptive Scheduling
C. Hand Shaking
D. Highest Penalty Ratio Next

ugcnetcse-dec2013-paper2 operating-system unix

Answer key

35.52.6 Unix: UGC NET CSE | December 2013 | Part 3 | Question: 67

In Unix, how do you check that two given strings a and b are equal?

A.
```
test $a -eq $b
```
B.
```
test $a -equal $b
```
C.
```
test $a = $b
```
D.
```
sh -C test $a == $b
```

ugcnetcse-dec2013-paper3 operating-system unix non-gatecse

Answer key

35.52.7 Unix: UGC NET CSE | December 2013 | Part 3 | Question: 73

Which statement is not true about process O in the Unix operating system ?

A. Process O is called init process.
B. Process O is not created by fork system call.
C. After forking process 1, process O becomes swapper process.
D. Process O is a special process created when system boots

ugcnetcse-dec2013-paper3 operating-system unix

Answer key

35.52.8 Unix: UGC NET CSE | December 2013 | Part 3 | Question: 74

Which of the following commands would return process_id of sleep command?

A.
```
Sleep 1 and echo $
```
B.
```
Sleep 1 and echo $#
```
C.
```
Sleep 1 and echo $*
```
D.
```
Sleep 1 and echo $!
```

ugcnetcse-dec2013-paper3 operating-system non-gatecse unix

Answer key

35.52.9 Unix: UGC NET CSE | December 2015 | Part 2 | Question: 30

In Unix, the login prompt can be changed by changing the contents of the file_____

A. contrab
B. init
C. gettydefs
D. inittab

ugcnetcse-dec2015-paper2 operating-system unix

Answer key

35.52.10 Unix: UGC NET CSE | December 2015 | Part 3 | Question: 56

In Unix, the command to enable execution permission for file "mylife" by all is_____

A. Chmod ugo + X mylife
B. Chmod a + X mylife
C. Chmod + X mylife
D. All of the above

ugcnetcse-dec2015-paper3 unix operating-system non-gatecse

Answer key

35.52.11 Unix: UGC NET CSE | December 2015 | Part 3 | Question: 57

What will be the output of the following Unix command?

$ rm chap0[1-3]

A. `Remove file $\text{chap0[1-3]}$`

B. `Remove file $\text{chap01, chap02, chap03}$`

C. `Remove file $\text{chap [1-3]}$`

D. `None of the above`

ugcnetcse-dec2015-paper3 unix operating-system non-gatecse

Answer key

35.52.12 Unix: UGC NET CSE | January 2017 | Part 3 | Question: 51

Match the following for Unix file system :

	List-I		List-II
a.	Boot block	i.	Information about file system, free block list, free inode list etc.
b.	Super block	ii.	Contains operating system files as well as program and data files created by users.
c.	Inode block	iii.	Contains boot program and partition table.
d.	Data block	iv.	Contains a table for every file in the file system. Attributes of files are stored here.

Codes :

A. a-iii, b-i, c-ii, d-iv

B. a-iii, b-i, c-iv, d-ii

C. a-iv, b-iii, c-ii, d-i

D. a-iv, b-iii, c-i, d-ii

ugcnetcse-jan2017-paper3 operating-system unix

Answer key

35.52.13 Unix: UGC NET CSE | January 2017 | Part 3 | Question: 54

Which statements is not correct about "init" process in Unix?

A. It is generally the parent of the login shell.
B. It has PID 1.
C. It is the first process in the system.
D. Init forks and execs a 'getty' process at every port connected to a terminal.

ugcnetcse-jan2017-paper3 operating-system unix

Answer key

35.52.14 Unix: UGC NET CSE | July 2016 | Part 3 | Question: 49

In UNIX , processes that have finished execution but have not yet had their status collected are known as ________

A. Sleeping processes

B. Stopped Processes

C. Zombie Process

D. Orphan Processes

ugcnetcse-july2016-paper3 operating-system unix

Answer key

35.52.15 Unix: UGC NET CSE | July 2016 | Part 3 | Question: 50

In UNIX OS , when a process creates a new process using the fork() system call , which of the following state is shared between the parent process and child process

A. Heap
B. Stack
C. Shared memory segments
D. Both heap and Stack

ugcnetcse-july2016-paper3 operating-system unix

Answer key ☟

35.52.16 Unix: UGC NET CSE | July 2016 | Part 3 | Question: 51

Which of the following information about the UNIX file system is not correct?

A. Super block contains the number of i-nodes, the number of disk blocks, and the start of the list of free disk blocks
B. An i-node contains accounting information as well as enough information to locate all the disk blocks that holds the file's data
C. Each i-node is 256-bytes long
D. All the files and directories are stored in data blocks

ugcnetcse-july2016-paper3 operating-system unix

Answer key ☟

35.52.17 Unix: UGC NET CSE | June 2013 | Part 2 | Question: 50

The mv command changes

A. the inode
B. the inode-number
C. the directory entry
D. both the directory entry and the inode

ugcnetcse-june2013-paper2 unix

Answer key ☟

35.52.18 Unix: UGC NET CSE | June 2014 | Part 3 | Question: 37

__________ maintains the list of free disk blocks in the Unix file system.

A. I-node
B. Boot block
C. Super block
D. File allocation table

ugcnetjune2014iii operating-system unix

Answer key ☟

35.52.19 Unix: UGC NET CSE | June 2014 | Part 3 | Question: 46

The output generated by the LINUX command :

$ seq 1 2 10

will be

A. 1 2 10
B. 1 2 3 4 5 6 7 8 9 10
C. 1 3 5 7 9
D. 1 5 10

ugcnetjune2014iii operating-system unix

Answer key ☟

35.52.20 Unix: UGC NET CSE | Junet 2015 | Part 2 | Question: 40

The Unix Kernel maintains two key data structures related to processes, the process table and the user structure. Which of following information is not the part of user structure?

A. File descriptor table
B. System call state
C. Scheduling parameters
D. Kernel stack

ugcnetcse-june2015-paper2 unix

Answer key

35.52.21 Unix: UGC NET CSE | Junet 2015 | Part 3 | Question: 54

A unix file may be of the type

A. Regular file
B. Directory file
C. Device file
D. Any one of the above

ugcnetcse-june2015-paper3 operating-system unix

Answer key

35.52.22 Unix: UGC NET CSE | Junet 2015 | Part 3 | Question: 73

Match the following for UNIX system calls :

	List-I		List-II
(a)	exec	(i)	Creates new process
(b)	brk	(ii)	Invokes another program overlaying memory space with a copy of an executable file.
(c)	wait	(iii)	To increase or decrease the size of data region
(d)	fork	(iv)	A process synchronizes with termination of child process

Codes :

A. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
B. (a)-(iii), (b)-(ii), (c)-(iv), (d)-(i)
C. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
D. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)

ugcnetcse-june2015-paper3 operating-system unix

Answer key

35.52.23 Unix: UGC NET CSE | September 2013 | Part 2 | Question: 47

Assume that an implementation of Unix operating system uses i-nodes to keep track of data blocks allocated to a file. It supports 12 direct block addresses, one in direct block address and one double indirect block address. The file system has 256 bytes block size and 2 bytes for disk block address. The maximum possible size of a file in this system is

A. 16 MB
B. 16 KB
C. 70 KB
D. 71 KB

ugcnetsep2013ii operating-system unix

Answer key

35.52.24 Unix: UGC NET CSE | September 2013 | Part 3 | Question: 74

Match the following for Windows Operating System :

(a)	Hardware abstration Layer	(i)	Starting all processes, emulation of different operating system,security functions, transform character based applications to graphical representation
(b)	Kernel	(ii)	Export a virtual memory interface, support for symmetric multiprocessing, administration, details of mapping memory, configuring I/O buses, setting up DMA
(c)	Executive	(iii)	Thread scheduling, interrupt and exception handling,recovery after power failure
(d)	Win32 subsystem	(iv)	Objet manager, virtual memory manager, process manager, plug-and-Play and power manager

Codes :

A. (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)
B. (a)-(iv), (b)-(iii), (c)-(ii), (d)-(i)
C. (a)-(ii), (b)-(iii), (c)-(iv), (d)-(i)
D. (a)-(iii), (b)-(ii), (c)-(i), (d)-(iv)

ugcnetcse-sep2013-paper3 operating-system unix

Answer key

35.53 Virtual Memory (7)

35.53.1 Virtual Memory: UGC NET CSE | December 2008 | Part 2 | Question: 42

If the executing program size is greater than the existing RAM of a computer, it is still possible to execute the program if the OS supports :

A. multitasking
B. virtual memory
C. paging system
D. none of the above

ugcnetcse-dec2008-paper2 operating-system virtual-memory

Answer key

35.53.2 Virtual Memory: UGC NET CSE | December 2013 | Part 3 | Question: 70

Translation Look-aside Buffer (TLB) is

A. a cache-memory in which item to be searched is compared one-by-one with the keys
B. a cache-memory in which item to be searched is compared with all the keys simultaneously
C. An associative memory in which item to be searched is compared one-by-one with the keys
D. An associative memory in which item to be searched is compared with all the keys simultaneously

ugcnetcse-dec2013-paper3 operating-system cache-memory virtual-memory

Answer key

35.53.3 Virtual Memory: UGC NET CSE | December 2015 | Part 2 | Question: 26

A virtual memory has a page size of 1K words. There are eight pages and four blocks. The associative memory page table contains the following entries :

Page	Block
0	3
2	1
5	2
7	0

Which of the following list of virtual addresses (in decimal) will not cause any page fault if referenced by the CPU?

A. $1024, 3072, 4096, 6144$
B. $1234, 4012, 5000, 6200$
C. $1020, 3012, 6120, 8100$
D. $2021, 4050, 5112, 7100$

ugcnetcse-dec2015-paper2 operating-system virtual-memory

Answer key

35.53.4 Virtual Memory: UGC NET CSE | January 2017 | Part 3 | Question: 2

Which of the following is incorrect for virtual memory?

A. Large programs can be written
B. More I/O is required
C. More addressable memory available
D. Faster and easy swapping of process

ugcnetcse-jan2017-paper3 operating-system virtual-memory

Answer key

35.53.5 Virtual Memory: UGC NET CSE | June 2005 | Part 2 | Question: 36

Moving Process from main memory to disk is called :

A. Caching
B. Termination
C. Swapping
D. Interruption

ugcnetcse-june2005-paper2 operating-system virtual-memory

Answer key

35.53.6 Virtual Memory: UGC NET CSE | June 2014 | Part 2 | Question: 31

In a paged memory management algorithm, the hit ratio is 70%. If it takes 30 nanoseconds to search Translation Look-aside Buffer (TLB) and 100 nanoseconds (ns) to access memory, the effective memory access time is

A. 91 ns
B. 69 ns
C. 200 ns
D. 160 ns

ugcnetcse-june2014-paper2 operating-system virtual-memory

Answer key

35.53.7 Virtual Memory: UGC NET CSE | June 2019 | Part 2 | Question: 47

The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by

A. page size
B. physical size of memory
C. the instruction set architecture
D. number of processes in memory

ugcnetcse-june2019-paper2 virtual-memory paging

Answer key

35.54 Vsam (1)

35.54.1 Vsam: UGC NET CSE | December 2010 | Part 2 | Question: 36

The dynamic allocation of storage areas with VSAM files is accomplished by

A. Hashing
B. Control splits
C. Overflow areas
D. Relative recoding

ugcnetcse-dec2010-paper2 operating-system vsam

Answer key

35.55 Windows (5)

35.55.1 Windows: UGC NET CSE | December 2013 | Part 3 | Question: 68

In windows 2000 operating system all the processor-dependent code is isolated in a dynamic link library called

A. NTFS file system
B. Hardware abstraction layer
C. Microkernel
D. Process Manger

ugcnetcse-dec2013-paper3 operating-system windows non-gatecse

Answer key

35.55.2 Windows: UGC NET CSE | December 2014 | Part 3 | Question: 73

Which of the following versions of Windows O.S. contain built-in partition manager which allows us to shrink and expand pre-defined drives ?

A. Windows Vista
B. Windows 2000
C. Windows NT
D. Windows 98

ugcnetcse-dec2014-paper3 operating-system windows non-gatecse

Answer key

35.55.3 Windows: UGC NET CSE | December 2015 | Part 3 | Question: 55

The character set used in Windows 2000 operating system is

A. 8 bit ASCII
B. Extended ASCII
C. 16 bit UNICODE
D. 12 bit UNICODE

ugcnetcse-dec2015-paper3 windows operating-system non-gatecse

Answer key

35.55.4 Windows: UGC NET CSE | June 2013 | Part 3 | Question: 66

The versions of windows operating system like Windows XP and Window Vista uses following file system:

A. FAT-16
B. FAT-32
C. NTFS (NT File System)
D. All of the above

ugcnetcse-june2013-paper3 operating-system windows

Answer key

35.55.5 Windows: UGC NET CSE | September 2013 | Part 3 | Question: 73

The portion of Windows 2000 operating system which is not portable is

A. processor management
B. user interface
C. device management
D. virtual memory management

ugcnetcse-sep2013-paper3 operating-system windows

Answer key

Answer Keys

35.0.1	C	35.0.2	A	35.0.3	A	35.0.4	A	35.0.5	A
35.0.6	A	35.0.7	C	35.0.8	A	35.0.9	C	35.0.10	B
35.0.11	C	35.0.12	B	35.0.13	B	35.0.14	C	35.0.15	X
35.0.16	D	35.0.17	C	35.0.18	Q-Q	35.0.19	Q-Q	35.0.20	N/A
35.0.21	Q-Q	35.0.22	B	35.0.23	Q-Q	35.0.24	Q-Q	35.0.25	Q-Q
35.0.26	Q-Q	35.0.27	X	35.0.28	B	35.0.29	Q-Q	35.0.30	Q-Q
35.0.31	Q-Q	35.0.32	B	35.0.33	C	35.0.34	B	35.0.35	X
35.0.36	N/A	35.0.37	Q-Q	35.0.38	Q-Q	35.0.39	Q-Q	35.0.40	Q-Q
35.0.41	Q-Q	35.0.42	Q-Q	35.0.43	Q-Q	35.0.44	Q-Q	35.0.45	Q-Q
35.0.46	Q-Q	35.0.47	B	35.0.48	Q-Q	35.0.49	Q-Q	35.1.1	C
35.2.1	Q-Q	35.2.2	Q-Q	35.2.3	A	35.3.1	B	35.4.1	C
35.5.1	C	35.6.1	Q-Q	35.6.2	Q-Q	35.6.3	B	35.6.4	Q-Q
35.6.5	Q-Q	35.6.6	D	35.6.7	A	35.6.8	Q-Q	35.6.9	A
35.7.1	D	35.8.1	C	35.9.1	A	35.10.1	A	35.10.2	B
35.10.3	C	35.10.4	C	35.10.5	B	35.10.6	C	35.10.7	B
35.10.8	A	35.10.9	A	35.10.10	D	35.11.1	B	35.12.1	C
35.12.2	X	35.12.3	D	35.12.4	B	35.12.5	C	35.12.6	B
35.12.7	D	35.12.8	C	35.12.9	B	35.13.1	B	35.14.1	Q-Q
35.15.1	B	35.16.1	Q-Q	35.17.1	Q-Q	35.17.2	C	35.17.3	X
35.17.4	B	35.17.5	B	35.17.6	A	35.17.7	B	35.18.1	Q-Q

35.19.1	Q-Q	35.20.1	B	35.21.1	Q-Q	35.22.1	A	35.22.2	C
35.23.1	B	35.23.2	C	35.24.1	A	35.25.1	Q-Q	35.26.1	Q-Q
35.26.2	Q-Q	35.26.3	Q-Q	35.26.4	X	35.26.5	A	35.26.6	B
35.26.7	C	35.26.8	C	35.27.1	A	35.28.1	A	35.29.1	D
35.30.1	C	35.31.1	C	35.31.2	D	35.31.3	Q-Q	35.32.1	Q-Q
35.32.2	Q-Q	35.32.3	D	35.32.4	C	35.32.5	B	35.32.6	A
35.32.7	B	35.32.8	D	35.32.9	D	35.33.1	A	35.34.1	B
35.34.2	B	35.35.1	Q-Q	35.35.2	Q-Q	35.36.1	Q-Q	35.36.2	Q-Q
35.36.3	Q-Q	35.36.4	Q-Q	35.36.5	Q-Q	35.36.6	B	35.36.7	B
35.36.8	D	35.36.9	Q-Q	35.36.10	C	35.36.11	B	35.36.12	Q-Q
35.36.13	Q-Q	35.36.14	Q-Q	35.36.15	A	35.36.16	D	35.36.17	B
35.36.18	D	35.36.19	B	35.36.20	B	35.36.21	B	35.36.22	Q-Q
35.36.23	C	35.36.24	A	35.36.25	A	35.37.1	Q-Q	35.37.2	Q-Q
35.38.1	Q-Q	35.39.1	A	35.39.2	B	35.40.1	Q-Q	35.41.1	Q-Q
35.41.2	Q-Q	35.41.3	B	35.41.4	Q-Q	35.42.1	B	35.43.1	Q-Q
35.43.2	A	35.43.3	D	35.44.1	D	35.44.2	C	35.44.3	D
35.44.4	B	35.44.5	B	35.44.6	C	35.44.7	B	35.45.1	Q-Q
35.46.1	D	35.47.1	D	35.48.1	A	35.48.2	C	35.49.1	D
35.50.1	B	35.50.2	B	35.50.3	A	35.50.4	C	35.51.1	C
35.52.1	3	35.52.2	Q-Q	35.52.3	B	35.52.4	A	35.52.5	B
35.52.6	C	35.52.7	A	35.52.8	D	35.52.9	C	35.52.10	B
35.52.11	B	35.52.12	B	35.52.13	C	35.52.14	C	35.52.15	C
35.52.16	C	35.52.17	C	35.52.18	C	35.52.19	C	35.52.20	C
35.52.21	D	35.52.22	A	35.52.23	X	35.52.24	C	35.53.1	Q-Q
35.53.2	D	35.53.3	C	35.53.4	B	35.53.5	Q-Q	35.53.6	D
35.53.7	C	35.54.1	Q-Q	35.55.1	B	35.55.2	A	35.55.3	C
35.55.4	D	35.55.5	D						